Transportation Research Part C 191 (2026) 105813



Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc





A stochastic delay model for signal-free intersections

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ARTICLE INFO

Keywords:
Signal-free junction
Stochastic modeling
Probability of conflict
Autonomous vehicles
Conflict networks
Queue theory

ABSTRACT

This study presents a stochastic delay model that treats signal-free intersections as systems of servers where the probability of conflict between vehicles determines delay. Rather than relying on predetermined major-minor hierarchies, the model builds on interaction between any two approaching vehicles attempting to occupy the same conflict zone simultaneously. This conflict-probability framework applies across many traffic scenarios, from traditional unsignalized multi-modal intersections, merging sections, and roundabouts to signal-free intersections for autonomous vehicles (AVs). The model employs queuing theory formulations where service time depends on conflict probability and clearance time for conflict zones. The model extends to multi-movement intersections through dependent and independent conflict zone networks and accommodates time-varying demand through dynamic parameter adaptation. Theoretical analysis reveals that equal inflows maximize delays due to peak conflict probabilities. Validation using real-world trajectory datasets demonstrates improvements over Highway Capacity Manual and Aastroads methods across diverse scenarios. Furthermore, the proposed delay model achieves mean absolute errors of 4.6 to 10.9 s using simulation of signal-free intersections for AVs under first-come first-served and communication-free distributed control algorithm protocols.

1. Introduction

Delay is a fundamental performance measure at intersections, capturing the additional time vehicles experience beyond ideal conditions. For human drivers, established traffic rules at intersections create a hierarchical priority mechanism that governs the characteristics of delay. Yet the emergence of autonomous vehicles (AVs) offers opportunities to redesign intersection control measures transcending traditional rule-based and priority mechanisms such as traffic signals or stop signs. This paper proposes a new stochastic delay model for signal-free intersections, ranging from common unsignalized intersections, roundabouts, merging and weaving sections, to signal-free intersections for AVs.

Delays at intersections arise from conflicts between vehicles attempting to occupy the same physical space during overlapping time intervals. Traditional delay estimation methods classify traffic streams into predetermined major and minor movements, estimating delays based on the minor stream's ability to find acceptable gaps in the major stream. These approaches maintain the assumption of fixed priority relationships that dictate which vehicles yield to others. This hierarchical framework presents limitations when applied to intersections where priority relationships are symmetric, dynamic, or non-existent.

On the other hand, the conflict-based perspective, that we proposed in this paper, focuses on the probability that vehicles from different approaches will simultaneously require access to intersection conflict zones. When two vehicles arrive such that their paths would intersect in both space and time, a conflict occurs, necessitating that one vehicle yields while the other proceeds. This yielding

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https://doi.org/10.1016/j.trc.2026.105813
Received 1 February 2026; Received in revised form 5 June 2026; Accepted 15 June 2026
Available online 30 June 2026
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behavior generates delay. The magnitude and distribution of delays depend on the likelihood of conflicts occurring and the time required to resolve them.

This study presents a general and stochastic delay estimation model that treats intersections as systems of conflict zones where vehicles compete for spatial-temporal resources. The model derives delay estimates from the probability of conflicts occurring between approaching vehicles, incorporating both the stochastic nature of arrivals and the time required for conflict resolution. The model extends from basic two-stream interactions to complex multi-movement networks, accommodates time-varying demand through dynamic parameter evolution, and provides analytical tools for both traditional unsignalized intersection and the those equipped with AVs coordination protocols.

1.1. Evolution of unsignalized intersection delay models

Over the past several decades, a wide array of analytical and empirical models have been developed to estimate delays at unsignalized intersections, integrating gap-acceptance and queueing processes.

Early Gap-Acceptance Models (1950s-1970s). One of the earliest analytical treatments of unsignalized intersection delay applied gap-acceptance theory to model a minor-road queue accepting gaps in a major-road traffic stream (Tanner, 1962). Tanner's model assumed a constant critical gap and Poisson arrivals on the major road, yielding a formula for average minor-stream delay as a function of conflicting flow rate. This laid the groundwork for subsequent delay models and is reflected in later practical guides such as the *Highway Capacity Manual* (HCM, Transportation Research Board, 1965), which adopted gap-acceptance concepts for capacity and delay estimation at stop-controlled intersections.

Early formulations using gap-acceptance provided reasonable estimates when traffic flows were low to moderate, but they struggled as conditions approached saturation (Kimber and Hollis, 1977). Recognizing this, researchers in the 1970s began to augment pure gap-acceptance theory with empirical corrections and queueing considerations. For instance, Kimber and Hollis (1977) developed one of the first comprehensive delay models for two-way stop-controlled intersections, relating average minor-road delay to the opposing traffic intensity. Their work, based on empirical observations in the UK, demonstrated that delay increases nonlinearly with the degree of conflict on the major road.

Queueing-Theoretic Models (1980s-1990s). By the 1980s, researchers increasingly viewed the delay problem through the lens of queueing theory. Troutbeck (1986) derived an analytical delay formula incorporating the minor approach's degree of saturation and an explicit term for queueing delay. Troutbeck's model introduced a 'queueing factor' that quantifies how delay escalates as the minor-road flow approaches its capacity, effectively blending gap-acceptance delay (experienced at low flows) with queueing delay (dominant at high flows). This improved the capture of delays under heavier traffic and influenced later editions of the HCM and other guidelines.

The early 1990s saw refinements to disentangle the components of delay. Heidemann (1991) and Kyte et al. (1991) each proposed methods to decompose total delay into two parts - (i) *service delay*, the time a vehicle waits for an acceptable gap (gap-acceptance delay), and (ii) *queueing delay*, the time spent waiting in a queue behind other vehicles that are also delayed. By linking these components to the traffic flows of both the minor approach and the conflicting major streams, their models provided insight into how major-road flow influences minor-road service times and how minor-road demand builds up queues. These formulations matched observed behavior more accurately than earlier models, especially in moderate traffic conditions where both gap-acceptance and queueing contribute appreciably to delay.

At the same time, researchers began to experiment with more general queueing models, for example, treating the minor approach as an M/M/1 queue (a single-server queue with Poisson arrivals and exponential service times) or variations thereof, to derive delay formulas. Under certain assumptions, the unsignalized intersection can indeed be approximated by an M/M/1 or M/G/1 queue with an initial headway distribution accounting for gap-acceptance. Daganzo (1977) formalized some of these ideas, describing the minor-stream delay problem in terms of an M/G/1 queue with an exceptional first service to represent the first vehicle's delay waiting for a gap. More recently, analytical delay frameworks based on shockwave theory have extended intersection delay modeling to minimize delay variability and spillback probability at undersaturated intersections (Mohajerpoor et al., 2019). This queueing perspective helped bridge the gap between gap-acceptance models and classical queueing theory results.

Combined Gap-Acceptance and Hybrid Models (late 1990s). As empirical evidence accumulated, it became clear that no single approach (pure gap-acceptance or pure queueing) sufficed across all traffic regimes. This led to the development of hybrid models that dynamically combine the two. For instance, Al-Omari and Benekohal (1999) proposed a hybrid delay model for two-way stop control that uses gap-acceptance theory to estimate delay under unsaturated conditions and then transitions to a queueing-based formulation as the minor approach becomes saturated. Such models acknowledge that when minor-road flow is low, delays are dominated by the process of finding an acceptable gap (as Tanner's model suggests); but when the minor-road demand exceeds what the available gaps can serve, a queue forms and additional delay accrues following queueing theory.

Another contribution was the comparison of prevailing models by Akcelik et al. (1998), who evaluated three different delay models (The HCM method, a gap-acceptance model, and a queueing model) against observed data. They found that while all models performed reasonably well in their intended range, discrepancies arose in over-saturated conditions. Notably, the simple exponential gap-acceptance formulas (older HCM versions) tended to underestimate delay once volume-to-capacity ratios approached 1.0, because they did not fully account for the compounding effect of queues.

These studies from the late 1990s underscored the importance of handling over-saturation explicitly, when demand exceeds capacity, delays increase sharply and can even become theoretically unbounded as queues grow without limit. Some analytical models began to borrow techniques from signalized intersection analysis to handle over-saturation. The 2000 HCM (Transportation Research Board,

2000), for example, incorporated an incremental delay term for two-way stop control, analogous to Webster's model for signalized intersections, which adds an over-saturation delay component once a critical flow is exceeded (Webster, 1958). Webster's original delay formula for signalized intersections separated delay into uniform delay (low arrivals) and an overflow delay (over-saturation); similarly, unsignalized delay models began to include an overflow component to maintain accuracy when minor-road demand is high. However, many of these adjustments were essentially deterministic and did not capture the stochastic nature of gap availability and queue formation, prompting further research into more probabilistic solutions.

Modern Conflict–Based Approaches. Over the past two decades, analytical treatment of unsignalized intersections has shifted from a rigid "major-minor" hierarchy to models that analyze each conflict point individually. Brilon and Wu (2002) recast capacity estimation as a set of mutually exclusive probabilistic servers, yielding closed–form capacities based on the chance that a movement is blocked by its conflicting streams. Field validation confirmed that these conflict–probability formulations, and the queueing models built on them, replicate observed delays more accurately than classical gap–acceptance or the HCM equations as demand approaches saturation (Brilon, 2008). Nevertheless, the technique still treats the service rate as an exogenous constant, preventing the probability of service from adapting to real–time flow fluctuations. It also analyzes conflicts in isolation, effectively limiting the framework to a single minor stream opposed by predefined priority flows and thereby overlooking interactions among multiple conflict zones. Moreover, the formulation is calibrated to human–driver behavior under quasi–stationary demand and produces essentially unimodal delay predictions, so it cannot reproduce the secondary delay peaks that emerge when competing flows are similarly high. Although later studies (Chandra et al., 2009; Prasetijo et al., 2016) extended the conflict framework to all–way stop control, auxiliary lanes, and heterogeneous traffic conditions, by embedding vehicle–type and acceleration differences into the conflict–stream formulation, these adaptations inherit the dynamic, multi–zone, and technology–specific limitations that the stochastic conflict–probability model proposed in this paper is designed to overcome.

Connected and Autonomous Vehicle Applications. While traditional studies focused on estimating delays given a certain control regime, the 2010s saw a surge of research on novel control strategies enabled by connectivity and automation. These studies often emphasized delay minimization through control algorithms rather than developing new analytic delay models. For example, Levin et al. (2016) demonstrated that dynamically adjusting vehicle right-of-way at intersections can reduce average delays compared to static control.

Another line of research introduced explicit communication-based coordination at intersections. Dresner and Stone (2008) proposed a multi-agent reservation system in which each approaching vehicle communicates with a centralized intersection manager to request access to the intersection. If the requested slot is free of conflicts with previously scheduled vehicles, the reservation is granted and the vehicle can pass without stopping; otherwise, it must wait until a conflict-free slot is available. Building on Dresner and Stone (2008), Vasirani and Ossowski (2012) proposed a market-inspired auction-based mechanism for AIM, improving over simple FCFS through economic coordination at both single-intersection and network levels. Levin and Boyles (2016) extended AIM communication protocols to dynamic lane reversal, achieving significant travel time reductions on city networks. Knezevic and Stevanovic (2024) incorporated pedestrian operations into reservation-based control within the CADLARIC framework, demonstrating safety improvements over fully reservation-based control under high demand. Moreover, these methods do not directly yield simple delay formulas; they provide strategies to minimize delay, but the delay outcomes typically have to be evaluated via simulation or complex analysis.

Recognizing the gap in analytical delay modeling for autonomous intersections, Yu et al. (2019) applied queueing theory to analyze vehicle delay under FCFS control, classifying the intersection as an M/G/1 system where service time follows a binary distribution determined by safety gaps between same-stream and conflicting-stream vehicles. They claimed that conflict probability is maximized when traffic flows on all approaches are equal, though this finding was demonstrated through numerical examples and applies only to their specific FCFS queueing interpretation. Their formulation uses safety time gaps (γ^s for same-stream vehicles, γ^d for conflicting streams) and calculates expected service time as $\mathbb{E}[S] = \gamma^d - (\gamma^d - \gamma^s)\sum_j P_j^2$, where P_j represents the probability that a vehicle originates from stream j. Yao et al. (2025) extended queueing-based capacity analysis for reservation-based autonomous intersections to the full intersection level, explicitly modeling crossing, merging, diverging, and conflict-free interactions as an M/G/1 system. They show that treating an intersection as a single conflict zone underestimates capacity and propose calibrated models achieving errors below 4% across diverse lane configurations. Section 4.2 compares their theoretical predictions against the proposed conflict-based model.

1.2. Proposed conflict-probability model

Traditional gap-acceptance models for unsignalized intersections rely on predetermined major-minor hierarchies and cannot adapt to scenarios where priority relationships are symmetric or dynamically changing, such as in all-way stops or autonomous vehicle coordination.

In light of the above, this study presents a delay model for unsignalized intersections that addresses the shortcomings of traditional methods by explicitly focusing on conflict probabilities. Rather than classifying streams into major and minor, or assuming a predefined right-of-way rule, the model considers the probability that any two approaching vehicles will conflict (i.e. attempt to occupy the same space; the intersection's conflict zone; at the same time) and uses this to determine delay. If two vehicles on conflicting approaches arrive such that their crossing paths overlap in time, a conflict occurs and one vehicle must yield, incurring delay while the other is served. The core concept is to treat the intersection as a stochastic system of servers where a vehicle is served once it completes its entire movement through the intersection, having successfully traversed all conflict zones along its path.

We model this process using a stochastic queueing framework. The server in this analogy is the shared conflict zone that can only be occupied by one vehicle at a time for safety; the service time corresponds to the time a vehicle occupies the conflict zone; and customers are vehicles from all approaches combined. A key point is that the service rate in our model is not fixed but depends on the arrival rates of conflicting streams through the probability of conflict; heavy flow on one approach effectively reduces the service rate for vehicles on other approaches by increasing the likelihood that the conflict zone is occupied.

By deriving the probability of conflict, we can compute the expected delay for vehicles on each approach. This model is probabilistic at its core, in contrast to the deterministic gap-acceptance threshold of classical models. It captures the random nature of vehicle arrivals and conflicts on a vehicle-by-vehicle basis. Moreover, it can accommodate various control scenarios. In fact, if an intersection had a traditional priority rule, that simply emerges as a special case in our model where the conflict probability is almost always resolved in favor of the major road except when it is clear.

The Poisson arrival process represents the standard analytical framework in traffic engineering for modeling vehicle arrivals when individual vehicles arrive independently. This assumption underpins classical traffic flow theory (Haight and Jacobson, 1963), gap-acceptance models (Tanner, 1962), Highway Capacity Manual methodologies (Transportation Research Board, 2010), and analytical intersection delay formulations (Akcelik, 1981). The Poisson process is characterized by exponentially distributed inter-arrival times and provides analytical tractability for deriving closed-form expressions.

While real traffic may deviate from perfect Poisson arrivals due to platooning, upstream signal coordination, and driver heterogeneity, the Poisson approximation has proven robust for aggregate performance estimation across diverse applications (Albin, 1982). For our conflict-probability framework, the Poisson assumption enables analytical derivation of conflict probabilities and service time distributions, yielding tractable delay formulations. Section 3.5 presents empirical analysis demonstrating that despite diversity in actual headway distributions, the Poisson-based model maintains high predictive accuracy across real-world validation datasets.

The remainder of this paper is structured as follows. Section 2 develops the theoretical methodology, beginning with conflict probability derivation for two-stream interactions and extending to multi-zone networks with both independent and dependent conflict scenarios. The section establishes M/M/1 and M/G/1 queueing formulations, addresses non-stationary demand through dynamic parameter evolution, and provides mathematical analysis of asymptotic behavior and model generalization across transport modes. Section 3 presents empirical validation using real-world trajectory data from intersections (inD dataset), roundabouts (rounD dataset), and highway merging sections (exiD dataset), demonstrating improved performance over traditional gap-acceptance methods with improvement across diverse traffic scenarios. Section 4 validates the model through autonomous vehicle simulations under both first-come first-served (FCFS) and communication-free distributed control algorithm (CfDCA) protocols, confirming theoretical predictions across constant and time-varying demand scenarios. Sections 5 and 6 synthesize findings through discussion and conclusion.

2. Methodology

This study models delay using queueing theory, considering stochastic arrival flows and service rate dynamics. The service rate for a given intersection approach is influenced by the arrival rate of its conflicting approach, with the conflict probability formulated as a probabilistic interaction between two stochastic processes. For any two conflicting movements (an example is illustrated in Fig. 1), vehicles must sequentially share a conflict zone to navigate through the intersection, underscoring the central role of conflict resolution in delay dynamics.

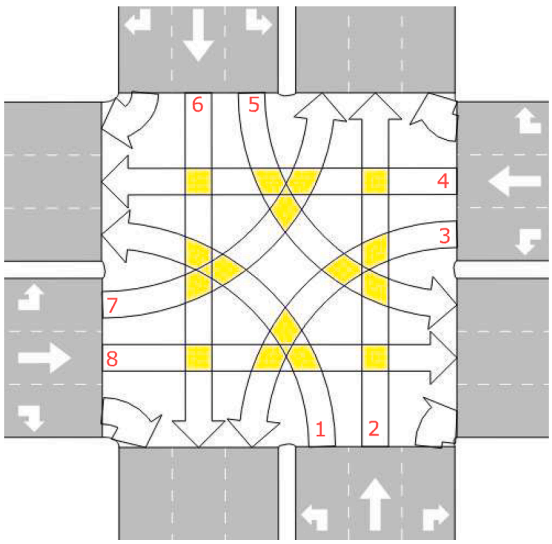


Fig. 1. Pairwise conflict zones inside an intersection.

When no conflict exists, the maximum service rate is determined by the minimum headway, $h_{\min}$, with the maximum capacity expressed as ${h_{\min}}^{-1}$. At low inflows, vehicles manage collision risks with minimal impact. However, as inflows rise, reduced headway increases conflict probability, causing delays that affect subsequent vehicles.

This section develops the conflict-based delay model through a progression from two-stream conflict probability derivation to multi-stream formulations. The analysis begins with basic conflict probability mechanics, evolves through service rate dynamics and queueing theory integration, extends to multi-zone networks via independent and dependent conflict formulations, and culminates with time-varying demand accommodation. The key parameters and symbols used throughout this paper are defined in Table 1.

Table 1
Table of nomenclature.

Symbol	Description	Unit
λ	General arrival rate	veh/s
λ_i	Arrival rate for approach i	veh/s
λ_k	Arrival rate for conflict zone k	veh/s
λ_m	Arrival rate for movement m	veh/s
$\lambda_{i_k}, \lambda_{j_k}$	Arrival rates for movements i_k and j_k in zone k	veh/s
$\lambda_{\text{veh}}, \lambda_{\text{ped}}$	Vehicle and pedestrian arrival rates	veh/s, ped/s
$\lambda_m(t), \lambda_k(t)$	Time-dependent arrival rates	veh/s
μ	Service rate	veh/s
ρ	Traffic intensity	–
ρ_k, ρ_m	Traffic intensity for zone k, movement m	–
$\rho(t), \rho_k(t)$	Time-dependent traffic intensity	–
γ^s	Safety time gap between vehicles from the same stream	s
γ^d	Safety time gap between vehicles from conflicting streams	s
a_i	Free-vehicle fraction for stream i, $e^{-\lambda_i h_{\min}}$	–
a_{i_k}, a_{j_k}	Free-vehicle fractions for movements i_k and j_k in zone k	–
C	Total flow constant	veh/s
C_m^2	Coefficient of variation squared for movement m	–
D_k, D_m	Delay for zone k, movement m	s
D_{total}	Total cumulative delay	s
$\mathbb{E}[D]$	Expected delay	s
$\mathbb{E}[D_k], \mathbb{E}[D_m]$	Expected delay for zone k, movement m	s
$\mathbb{E}[D]_{\text{M/M/1}}$	Expected delay using M/M/1 formulation	s
$\mathbb{E}[D]_{\text{M/G/1}}$	Expected delay using M/G/1 formulation	s
$\mathbb{E}[D_k]_{\text{det}}$	Expected deterministic delay for zone k	s
$\mathbb{E}[D_{\text{system}}]$	System-level expected delay	s
$\mathbb{E}[S]$	Expected service time	s
$\mathbb{E}[S_k], E[S_m]$	Expected service time for zone k, movement m	s
$\mathbb{E}[S^2]$	Second moment of service time	s^2
$\mathbb{E}[W]$	Expected waiting time	s
K	Set of conflict zones encountered by movement	–
P_{conflict}	Probability of conflict	–
P_m	Movement-level conflict probability	–
$P_{m,k}$	Probability of conflict for movement m at zone k	–
$P_{\text{conflict}}^{(k)}$	Conflict probability for zone k	–
$P_{\text{conflict}}^{\text{ped-veh}}$	Pedestrian-vehicle conflict probability	–
P_j	Probability that an approaching vehicle is from stream j	–
$Q_k, Q_k(t)$	Queue length for zone k	veh
T	Critical gap	s
T_0	Follow-up time	s
T_i	Time-to-intersection for vehicle i	s
$h_{\min}$	Minimum headway	s
i_k, j_k	Movement indices for conflict zone k	–
k	Conflict zone index	–
m	Movement index	–
n	Number of movements	–
t	Time variable	s
t_{clear}	Conflict zone clearance time	s
$t_{\text{clear,ped}}, t_{\text{clear,veh}}$	Conflict zone clearance times for pedestrians and vehicles	s
t_0, t_f	Start and end times of observation period	s
$v(t)$	Vehicle speed at time t	m/s
v_{free}	Free-flow speed	m/s
x, y	Arrival times for vehicles from different approaches	s

2.1. Probability of conflict and service rate

We assume vehicles enter the intersection from each approach following a Poisson distribution with an arrival rate λ_i, making the inter-arrival time t exponentially distributed. As vehicles from the two approaches arrive independently, the conflict probability

P_{conflict} is derived by integrating the joint PDF of inter-arrival times over possible values (Eq. (2)). The critical point is when the absolute difference in arrival times of two vehicles from the two approaches at the conflict zone inside the intersection is less than the safe average time interval for conflict zone usage when conflict possibility exists (t_{clear}). The term t_{clear} denotes the clearance time for the conflict zone, representing the duration required for a vehicle to traverse and vacate a conflict zone, plus the subsequent safety interval before the next vehicle's entry.

The arrival rate λ in the conflict probability formulation represents the effective arrival rate observed empirically from traffic counts, which inherently reflects the achievable flow rate given spacing constraints. When vehicles maintain minimum headway $h_{\min}$, the maximum achievable arrival rate is bounded by $\lambda_{\max} = 1/h_{\min}$. The model applies for $\lambda < \lambda_{\max}$, ensuring stable operations. The conflict probability (Eq. (2)) depends on arrival rates and conflict window duration t_{clear}, capturing the likelihood of simultaneous conflict zone requirements between streams.

Conflict zones and clearance time: The model employs conflict zones, spatial regions where vehicle paths from different movements overlap sufficiently that simultaneous occupancy would violate safety spacing–rather than exact trajectory points. This zone-based representation accommodates natural path variability in human driving (e.g., left-turning vehicles taking tighter or wider arcs), defining zones based on geometric path overlap with realistic lateral variation (Brilon and Miltner, 2005; Troutbeck and Brilon, 1997).

The clearance time t_{clear} represents the effective minimum time gap between conflicting vehicles using the same zone. This is not a fixed infrastructure property but an emergent characteristic depending on: driver/controller behavior, perception capabilities, coordination efficiency, speed profiles, traffic composition (including vulnerable road users), and geometric constraints. Different control strategies or driver populations exhibit different t_{clear} values because they operate with different behavioral characteristics–more efficient coordination maintains tighter temporal spacing (lower t_{clear}, lower delays). The parameter is calibrated from observed system performance rather than specified a priori.

This zone-based representation accommodates natural path variability in human driving. For example, left-turning vehicles may take tighter or wider arcs depending on speed, vehicle characteristics, and driver behavior, but the conflict zone encompasses this variation. The zone represents the spatial envelope where conflicts between specific movement pairs can occur, not a requirement for vehicles to follow predetermined trajectories. This abstraction is standard in conflict-based capacity analysis (Brilon and Miltner, 2005; Troutbeck and Brilon, 1997).

The inter-arrival headways of each stream are modelled by the Cowan M3 distribution (Cowan, 1975; Luttinen, 1999), which naturally arises from a Poisson process with rate λ_i subject to a minimum spacing $h_{\min}$. A proportion $a_i = e^{-\lambda_i h_{\min}}$ of vehicles are *free*, their Poisson-generated headway already exceeds $h_{\min}$, while the remaining proportion $1 - a_i$ are *bunched*, constrained to travel at exactly $h_{\min}$. By the memoryless property of the exponential distribution, free vehicles have headways $h_{\min} + \text{Exp}(\lambda_i)$ with the same rate parameter λ_i (Luttinen, 1999). The full M3 density is:

$$f_{T_i}(t) = (1 - a_i)\,\delta(t - h_{\min}) + a_i\,\lambda_i\,e^{-\lambda_i(t - h_{\min})}, \quad t \geq h_{\min}, \quad a_i = e^{-\lambda_i h_{\min}}. \tag{1}$$

Here $\delta(\cdot)$ denotes the Dirac delta function representing the point mass at $t = h_{\min}$, and the two components correspond to bunched vehicles (first term) and free vehicles (second term), with the total integrating to unity.

For vehicles arriving at times x and y from two conflicting approaches, both measured from the start of the observation period with minimum headway constraints, a conflict occurs when the absolute difference in their arrival times at the conflict zone is less than the safe clearance time ($|x - y| < t_{\text{clear}}$). The joint distribution of (x, y) under the M3 density (Eq. (1)) decomposes into four cases: both bunched, one bunched and one free, or both free. Cases involving at least one bunched vehicle are resolved analytically; the both-free case yields a double integral identical in structure to the pure shifted-exponential formulation, scaled by $a_1 a_2$. Integrating the joint probability density function over all arrival time combinations that satisfy this conflict condition yields:

$$\begin{aligned} P_{\text{conflict}} = {} & a_1\,a_2 \cdot \lambda_1\lambda_2 \int_{h_{\min}}^{\infty} e^{-\lambda_1(x - h_{\min})} \left(\int_{\max(h_{\min},\, x - t_{\text{clear}})}^{x + t_{\text{clear}}} e^{-\lambda_2(y - h_{\min})}\,\mathrm{d}y \right) \mathrm{d}x \\ & + (1 - a_1)(1 - a_2) + (1 - a_1)\,a_2\left(1 - e^{-\lambda_2 t_{\text{clear}}}\right) + a_1\,(1 - a_2)\left(1 - e^{-\lambda_1 t_{\text{clear}}}\right), \end{aligned} \tag{2}$$

which can be simplified as (derivation in Appendix A):

$$P_{\text{conflict}} = 1 - \frac{a_1[\lambda_1(1 - a_2) + \lambda_2]\,e^{-\lambda_1 t_{\text{clear}}} + a_2[\lambda_2(1 - a_1) + \lambda_1]\,e^{-\lambda_2 t_{\text{clear}}}}{\lambda_1 + \lambda_2}, \quad a_i = e^{-\lambda_i h_{\min}}. \tag{3}$$

Note that when $a_i \to 1$, Eq. (3) reduces to the simpler form:

$$P_{\text{conflict}} = 1 - \frac{\lambda_1 e^{-\lambda_2 t_{\text{clear}}} + \lambda_2 e^{-\lambda_1 t_{\text{clear}}}}{\lambda_1 + \lambda_2}, \tag{4}$$

which corresponds to a pure shifted-exponential arrival model with no bunching. Although Eq. (4) is a special case of Eq. (3), it remains a highly practical formulation: at low demand the bunching fraction $1 - a_i = 1 - e^{-\lambda_i h_{\min}}$ is negligible, making both equations nearly identical. Furthermore, since t_{clear} is calibrated to minimise prediction error, any systematic difference between the two formulations is absorbed into the calibrated value of t_{clear}, provided calibration and deployment conditions are comparable. Eq. (4) is therefore adequate whenever the model is calibrated and applied under similar demand conditions; Eq. (3) is preferred when bunching is non-negligible or when the model is calibrated at low demand and applied at moderate-to-high demand.

For the same physical intersection geometry, different control strategies or driver populations can exhibit different effective clearance times because they operate with different behavioral characteristics. More efficient coordination strategies maintain tighter temporal spacing, resulting in lower t_{clear} values and consequently lower delays. Similarly, the presence of vulnerable road users such

as bicycles increases effective clearance times as both the vulnerable users and motor vehicle drivers adopt more conservative conflict resolution behavior. The parameter is calibrated from observed system performance, revealing the effective clearance behavior rather than being specified a priori as an infrastructure constant.

With the total inflow constant, the equal inflow of movements, i.e., $\lambda_1 = \lambda_2$, results in the highest probability of conflict and consequently the highest delay. We analyze expected delays for different arrival rates, assuming a maximum service rate of 1 vehicle per second. Fig. B.8 shows results for scenarios where $\lambda_1 + \lambda_2$ is constant. The maximum delay is observed when inflow is equally distributed across lanes. This proposition is proven in Appendix B.

Having established the conflict probability formulation, we now examine how this probability drives service time and delay.

2.2. Service time and service rate dynamics

Traditional models often treat the service rate at intersections as a constant, determined by the control strategy (e.g., signal timing). This assumption simplifies the analysis and is suitable for controlled systems where external mechanisms dictate vehicle sequencing. However, in signal-free intersections, particularly under stochastic traffic conditions, the service rate depends on the probability of conflicts between vehicles. A constant service rate fails to capture the dynamics between arrival flows, vehicle interactions, and conflict resolution, which are central to the performance of such systems. Initially, we need to explain some of the fundamental queueing concepts:

Stochastic traffic conditions refer to the inherent randomness in vehicle arrivals, speeds, and gap acceptance behavior that characterizes real traffic systems. Unlike deterministic models that assume fixed, predictable vehicle patterns, stochastic models recognize that: (1) vehicle arrival times vary randomly around mean rates, (2) driver gap acceptance decisions involve probabilistic elements, (3) service times (time to traverse conflict zones) fluctuate based on conflict occurrences, and (4) queue lengths and delays exhibit random variability even under constant demand. This contrasts with deterministic approaches where vehicle trajectories and timing are perfectly predictable. The stochastic framework captures the reality that traffic systems operate under uncertainty, with individual vehicle behavior contributing to aggregate randomness in system performance. Queueing theory provides the mathematical tools to analyze expected performance metrics (mean delay, queue length) under these stochastic conditions (Gross, 2008).

Binary service time distribution: In our formulation, the service time for each vehicle follows a binary (two-point) distribution based on whether a conflict occurs. If a conflict is detected (with probability P_{conflict}), the vehicle must wait for clearance time t_{clear}. If no conflict occurs (with probability $1 - P_{\text{conflict}}$), the vehicle passes with only the minimum headway delay $h_{\min}$. This binary structure differs from continuous service time distributions assumed in many classical queueing models. The first and second moments of this distribution (Eqs. (3) and (5)) are used in subsequent queueing analysis.

M/M/1 queueing model: The notation M/M/1 refers to a single-server queue with Markovian (memoryless) arrivals, Markovian (exponential) service times, and one server. The first M indicates Poisson arrivals (exponentially distributed inter-arrival times), the second M indicates exponentially distributed service times, and 1 indicates a single server. For this system, the expected delay is given by Eq. (6), where $\rho = \lambda \mathbb{E}[S]$ is the traffic intensity. This formulation assumes exponential service times, which is a simplification for our binary service distribution but provides a baseline comparison (Gross, 2008).

M/G/1 queueing model: The notation M/G/1 indicates Markovian arrivals, General service time distribution, and one server. The M/G/1 model accommodates arbitrary service time distributions through the Pollaczek-Khinchine formula (Eq. (7)). This formulation explicitly accounts for service time variability through the second moment $\mathbb{E}[S^2]$. Since our binary service time distribution is not exponential, M/G/1 represents the theoretically appropriate framework (Kleinrock, 1975). The difference between M/M/1 and M/G/1 predictions reflects the impact of service time variability on delay, with M/G/1 providing more accurate results for non-exponential service distributions.

Recovery of stochastic arrival process: In independent conflict zone networks (e.g., roundabouts), vehicles negotiate conflict zones sequentially with sufficient spatial separation between zones. After clearing one conflict zone, vehicles accelerate back to normal speeds and spacing before encountering the next zone. This spatial separation allows the traffic stream to recover its stochastic arrival characteristics, meaning that the arrival process at the second conflict zone is statistically independent of the service process at the first zone.

This recovery enables treatment of each conflict zone as an independent M/M/1 or M/G/1 queue, with delays summing linearly across zones. In contrast, dependent networks (intersections) require vehicles to commit to traversing all conflict zones once entered, preventing recovery between zones and necessitating different analytical treatment where the movement-level conflict probability accounts for all zones collectively.

The concept of statistical independence between queueing stages through spatial separation is well-established in queueing network theory (Jackson, 1957; Kelly, 2011). For traffic applications, Brilon and Wu (2021) discuss this principle in the context of roundabout capacity analysis, where entry and circulating conflicts can be treated as sequential independent processes.

As defined in Section 2, t_{clear} is an effective emergent parameter calibrated per intersection. When no conflict arises, vehicles traveling in the same movement can maintain a minimum headway, $h_{\min}$, determined by their desired following behavior and safety constraints. Hence, the service time for a single vehicle reflects one of two scenarios; with probability P_{conflict}, the service time equals the safe clearance time t_{clear}, while otherwise it simplifies to $h_{\min}$. These scenarios can be combined into an expected service time $\mathbb{E}[S]$:

$$\mathbb{E}[S] = P_{\text{conflict}}\, t_{\text{clear}} + \left(1 - P_{\text{conflict}}\right) h_{\min}. \tag{5}$$

This captures both the conflict-driven delay and the uninterrupted flow conditions and indicates how increased conflict probability or longer safe clearance time for the conflict zone reduces intersection throughput and thus elevates overall delays.

Let traffic intensity ρ be defined as:

$$\rho = \lambda \mathbb{E}[S]. \tag{6}$$

This relationship applies to non-congested conditions where traffic intensity $\rho < 1$, with a single conflict zone. The binary service time distribution (conflict vs. non-conflict scenarios) enables direct calculation of the second moment $\mathbb{E}[S^2]$ required for M/G/1 analysis:

$$\mathbb{E}[S^2] = P_{\text{conflict}} t_{\text{clear}}^2 + (1 - P_{\text{conflict}}) h_{\text{min}}^2. \tag{7}$$

This formulation represents the system-level expected service time, where P_{conflict} captures the probability that any vehicle entering the system will encounter a conflicting situation requiring extended clearance time. The model does not track individual vehicle interactions but rather characterizes the aggregate service behavior of the intersection system under stochastic arrival conditions. The expected delay $E[D]$ comprises waiting time $E[W]$ and service time $E[S]$, with traffic intensity $\rho = \lambda E[S] < 1$ required for stable operations. For M/M/1, Little's Law gives $E[W] = E[S]/(1-\rho) - E[S]$, so $E[D]_{M/M/1} = E[S]/(1-\rho)$. For M/G/1, the Pollaczek-Khinchine formula (Kleinrock, 1975; Gross, 2008) gives $E[W] = \lambda E[S^2]/(2(1-\rho))$, so $E[D]_{M/G/1} = \lambda E[S^2]/(2(1-\rho)) + E[S]$, explicitly accounting for service time variability through the second moment $E[S^2]$ from Equation (5). Therefore, the expected delay using Eq. (8) for M/M/1 formulation $\mathbb{E}[D]_{\text{M/M/1}}$ (Exponential Service Assumption), and Eq. (9) for M/G/1 formulation $\mathbb{E}[D]_{\text{M/G/1}}$ (General Service Distribution) with the Pollaczek-Khinchine formula and the second moment from Eq. (7), including the service time are:

$$\mathbb{E}[D]_{\text{M/M/1}} = \mathbb{E}[W] + \mathbb{E}[S] = \frac{\rho}{1-\rho}\mathbb{E}[S] + \mathbb{E}[S] = \frac{1}{1-\rho}\mathbb{E}[S], \tag{8}$$

$$\mathbb{E}[D]_{\text{M/G/1}} = \mathbb{E}[W] + \mathbb{E}[S] = \frac{\lambda}{2(1-\rho)}\mathbb{E}[S^2] + \mathbb{E}[S]. \tag{9}$$

The binary service time distribution (conflict or no conflict) is not exponential, making M/G/1 the theoretically appropriate queueing framework. The M/G/1 formulation (Eq. (9)) explicitly accounts for service time variability through the second moment (Eq. (7)), capturing the impact of the binary service process on delay. We also present M/M/1 results (Eq. (8)) as a simplified baseline for comparison, though M/G/1 represents the more rigorous theoretical treatment.

The above single-zone formulation provides the foundation for understanding conflict-based delays, but real intersections involve multiple movements with complex interaction patterns. The extension to multi-zone analysis requires addressing how delays accumulate when a vehicle encounters multiple potential conflicts along its path. The answer depends critically on the spatial and temporal relationships between conflict zones, leading to two distinct analytical approaches.

2.3. Conflict-zone network and delay formulation

Real facilities involve multiple movements that create complex networks of conflict zones. Consider an intersection with n distinct movements where vehicles must traverse multiple conflict zones along their paths. Each conflict zone k represents a spatio-temporal region where conflicting movements cannot simultaneously operate safely.

When vehicles traverse multiple conflict zones, the spatial relationship between these zones determines how delays accumulate. At roundabouts, vehicles encounter sequential conflicts with sufficient separation to recover normal flow characteristics between zones, allowing delays to accumulate linearly across the traveled path. Conversely, at geometrically constrained intersections, vehicles must commit to complete path traversal upon initial zone entry, creating interdependent conflicts where the overall conflict probability reflects the likelihood of encountering any conflict during the entire maneuver. This operational distinction fundamentally governs whether delays aggregate through linear summation or require joint probabilistic analysis of the delay throughout the complete movement path.

The choice between independent and dependent formulations fundamentally affects the delay calculation methodology and determines the appropriate aggregation approach for network-level analysis.

2.3.1. Intersection-level delay aggregation

The delay aggregation method differs for independent versus dependent conflict zone networks. This approach obtains expected delays for individual movements, but these values lack physical meaning without knowledge of the intersection's operational rules. The meaningful metric emerges only through demand-weighted averaging across all movements, which yields the system-level expected delay.

Independent Conflict Zone Networks When conflict zones exhibit sufficient spatial separation to permit recovery of stochastic arrival processes, zone delays aggregate linearly. Let us denote K as the set of conflict zones encountered by movement m along its path through the intersection. Accordingly,

M/M/1 Formulation:

$$\mathbb{E}[D_m]_{\text{M/M/1}} = \sum_{k \in K} \mathbb{E}[D_k]_{\text{M/M/1}} = \sum_{k \in K} \left(\frac{\mathbb{E}[S_k]}{1-\rho_k} \right). \tag{10}$$

M/G/1 Formulation:

$$\mathbb{E}[D_m]_{\text{M/G/1}} = \sum_{k \in K} \mathbb{E}[D_k]_{\text{M/G/1}} = \sum_{k \in K} \left(\frac{\lambda_k}{2(1-\rho_k)} \mathbb{E}[S_k^2] + \mathbb{E}[S_k] \right). \tag{11}$$

Dependent Conflict Zone Networks In geometrically constrained environments where vehicles must commit to complete path traversal upon initial zone entry, the movement-level conflict probability becomes:

$$P_m = 1 - \prod_{k=1}^{K}(1 - P_{m,k}), \tag{12}$$

where $P_{m,k}$ is the probability of conflict for movement m at conflict k from Eq. (3), and P_m represents the probability of experiencing at least one conflict along the movement path. Eq. (12) follows from the complement rule applied to independent conflict events: the probability of no conflict at any zone along the path is $\prod_{k=1}^{K}(1 - P_{m,k})$, giving $P_m = 1 - \prod_{k=1}^{K}(1 - P_{m,k})$. The parameter t_{clear} represents an effective movement-level clearance time aggregated over all conflict types along the path, calibrated from observed delay data. Conflict-type-specific clearance times, as in Yao et al. (2025), are identified as a direction for future extension. The movement-level service characteristics are then updated in Eqs. (5) and (7) with P_m, yielding movement delays:

M/M/1 Dependent Formulation:

$$\mathbb{E}[D_m]_{\text{M/M/1}} = \frac{\mathbb{E}[S_m]}{1 - \lambda_m \mathbb{E}[S_m]}. \tag{13}$$

M/G/1 Dependent Formulation:

$$\mathbb{E}[D_m]_{\text{M/G/1}} = \frac{\lambda_m \mathbb{E}[S_m^2]}{2(1 - \lambda_m \mathbb{E}[S_m])} + \mathbb{E}[S_m]. \tag{14}$$

System Performance Aggregation

The overall system performance is characterized through flow-weighted delay averaging:

$$\mathbb{E}[D_{\text{system}}] = \frac{\sum_{m=1}^{n} \left(\lambda_m \mathbb{E}[D_m]\right)}{\sum_{m=1}^{n}(\lambda_m)}. \tag{15}$$

This formulation provides a framework for delay estimation across diverse intersection configurations under stable, low-demand conditions while maintaining mathematical tractability and computational efficiency. The approach assumes non-congested operations where $\rho_k < 1$ for all conflict zones and relies on steady-state queueing principles. For near-capacity conditions, over-saturated scenarios, and time-varying demand patterns that violate these assumptions, the extensions developed in the subsequent section provide the necessary analytical tools to handle complex operational regimes and non-stationary traffic conditions.

2.4. Non-stationary demand and dynamic service rate analysis

The conflict-based formulations established through the probability derivation, service rate dynamics, and network extensions assume steady-state conditions where arrival rates and conflict probabilities remain temporally constant. Real traffic systems, however, experience dynamic demand fluctuations through morning rush periods, incident-induced congestion, and operational modifications that create time-varying conditions requiring analytical extension.

Under non-stationary demand, both conflict probability $P_{\text{conflict}}(t)$ and service characteristics $\mathbb{E}[S](t)$ evolve continuously with temporal demand patterns. When instantaneous demand temporarily exceeds capacity ($\rho(t) \geq 1$), the stochastic delay formulations transition from probabilistic service variability to deterministic queue accumulation dynamics. The following analysis extends the conflict-based model to accommodate these temporal variations while preserving analytical tractability for practical implementation.

Under non-stationary conditions, let $\lambda_m(t)$ represent the instantaneous arrival rate for movement m at time t, where we assume t_{clear} and $h_{\min}$ remain constant parameters. The composite zone arrival rate becomes $\lambda_k(t) = \lambda_{i_k}(t) + \lambda_{j_k}(t)$ for conflict zone k involving movements i_k and j_k. The traffic intensity ($\rho_k(t)$) determines the instantaneous operational regime.

The fundamental recognition underlying this extension is that both conflict probability and service time evolve dynamically with demand patterns. The instantaneous conflict probability follows the established formulation with time-dependent parameters:

$$P_{\text{conflict}}^{(k)}(t) = 1 - \frac{a_{i_k}(t)\left[\lambda_{i_k}(t)(1 - a_{j_k}(t)) + \lambda_{j_k}(t)\right]e^{-\lambda_{i_k}(t)t_{\text{clear}}} + a_{j_k}(t)\left[\lambda_{j_k}(t)(1 - a_{i_k}(t)) + \lambda_{i_k}(t)\right]e^{-\lambda_{j_k}(t)t_{\text{clear}}}}{\lambda_{i_k}(t) + \lambda_{j_k}(t)}, \tag{16}$$

where $a_{i_k}(t) = e^{-\lambda_{i_k}(t)h_{\min}}$ and $a_{j_k}(t) = e^{-\lambda_{j_k}(t)h_{\min}}$. This yields time-dependent service characteristics that maintain the binary distribution structure while adapting to instantaneous demand conditions by the updates in Eqs. (5) and (7) with $P_{\text{conflict}}^{(k)}(t)$.

The conflict-based approach inherently provides system capacity through the effective service rate calculation, eliminating the need for separate capacity estimation procedures. As demand approaches capacity where $\rho_k(t) \to 1$, the stochastic delay formulations theoretically yield infinite expected delays, reflecting the infinite-time horizon assumption of steady-state queueing theory. However, such conditions represent brief transitional periods during demand fluctuations rather than sustained operational states in real traffic systems.

Under non-stationary demand, both conflict probability $P_{\text{conflict}}(t)$ and service characteristics $\mathbb{E}[S](t)$ evolve continuously with temporal demand patterns. The delay estimation employs a quasi-steady-state approximation: at each time instant, we assume the system has reached steady state given the current arrival rates and conflict probabilities, and apply the steady-state queueing formulas with instantaneous parameter values.

This approximation is valid when demand variations occur slowly relative to queueing dynamics, allowing the system to track the time-varying equilibrium. For traffic systems, this typically holds when demand changes occur over time scales of minutes rather than seconds, which is characteristic of real traffic patterns (morning/evening peaks, incident effects).

During periods when $\rho_k(t) < 1$, the quasi-steady-state formulations provide delay estimates Eqs. (17) and (18). These are steady-state queueing results applied with time-varying parameters, not true transient queueing solutions. The notation t indicates parameter evaluation time, not that the formulas account for transient queue dynamics.

$$\mathbb{E}[D_k]_{\mathrm{M/M/1}}(t) = \frac{\mathbb{E}[S_k](t)}{1-\rho_k(t)}, \quad \rho_k(t) < 1 \tag{17}$$

$$\mathbb{E}[D_k]_{\mathrm{M/G/1}}(t) = \frac{\lambda_k(t)\,\mathbb{E}[S_k^2](t)}{2(1-\rho_k(t))} + \mathbb{E}[S_k](t), \quad \rho_k(t) < 1 \tag{18}$$

The quasi-steady-state approximation has limitations. It does not capture transient effects when demand changes rapidly, nor does it account for queue buildup and dissipation dynamics during transitions between demand levels. True transient queueing analysis would require solving time-dependent differential or integral equations, which typically lack closed-form solutions and require numerical methods.

When demand temporarily exceeds capacity ($\rho_k(t) \geq 1$), the system behavior transitions to deterministic queue accumulation where delays are governed by queue clearance principles rather than stochastic service variability. The conflict-based service rate formulation remains valid, but delay calculation shifts to deterministic analysis. For vehicles arriving during over-saturated periods, the expected deterministic delay becomes Eq. (19).

$$\mathbb{E}[D_k]_{\mathrm{det}}(t) = Q_k(t)\mathbb{E}[S_k](t) + \mathbb{E}[S_k](t) = (Q_k(t)+1)\mathbb{E}[S_k](t), \tag{19}$$

where $Q_k(t)$ represents the instantaneous queue length and this formulation provides the expected delay for a newly arriving vehicle that joins the queue at time t. This vehicle-specific delay reflects the deterministic FCFS waiting process; the arriving vehicle waits for all $Q_k(t)$ vehicles ahead to be served, each requiring an average service time of $\mathbb{E}[S_k](t)$, followed by its own service duration.

The quasi-steady-state approximation employed here represents a pragmatic approach for time-varying demand analysis. Alternative methodologies exist in the traffic engineering literature for handling non-stationary conditions.

The Highway Capacity Manual (HCM) employs incremental delay formulations that gradually blend uniform delay and overflow delay components across the operational range. Similarly, Troutbeck's transition techniques for unsignalized intersections provide time-dependent solutions through continuous blending of gap-acceptance delay and queueing delay as demand varies. These approaches typically initiate transitions at low traffic intensities ($\rho \approx 0$) and gradually increase the overflow or queueing component as demand approaches capacity.

Our approach differs by maintaining a sharp transition at $\rho_{\mathrm{threshold}} = 0.95$. The rationale is that intersections operate in distinct physical regimes: stochastic conflict resolution dominates at low to moderate demand, while deterministic queue processing dominates during capacity exceedances. Maintaining the pure stochastic formulation throughout the stable regime ($\rho < 0.95$) preserves the accuracy of the conflict-probability framework, while the decisive switch to deterministic delay accounts for queue accumulation when it becomes the dominant mechanism.

The comparison between gradual blending (HCM/Troutbeck approach) and concentrated transition (our approach) reflects different modeling philosophies: gradual blending provides smooth mathematical transitions and accounts for intermediate regimes but requires calibration of blending functions across the full range, while concentrated transition maintains theoretical purity of each regime and simpler implementation but requires determination of appropriate threshold. Empirical validation suggests that for signal-free intersections analyzed in this study, the concentrated transition approach provides superior accuracy, as demonstrated by validation results (Table 5) and time-varying simulation performance (Fig. 7).

To handle the transition between stochastic and deterministic regimes, we implement a regime-switching approach with threshold traffic intensity $\rho_{\mathrm{threshold}} = 0.95$. At the transition point, we compute the stochastic delay $D_{\mathrm{stoch}}(\rho_{\mathrm{threshold}})$ and combine it with deterministic queue processing. The deterministic formulation for $\rho_k(t) \geq \rho_{\mathrm{threshold}}$ becomes:

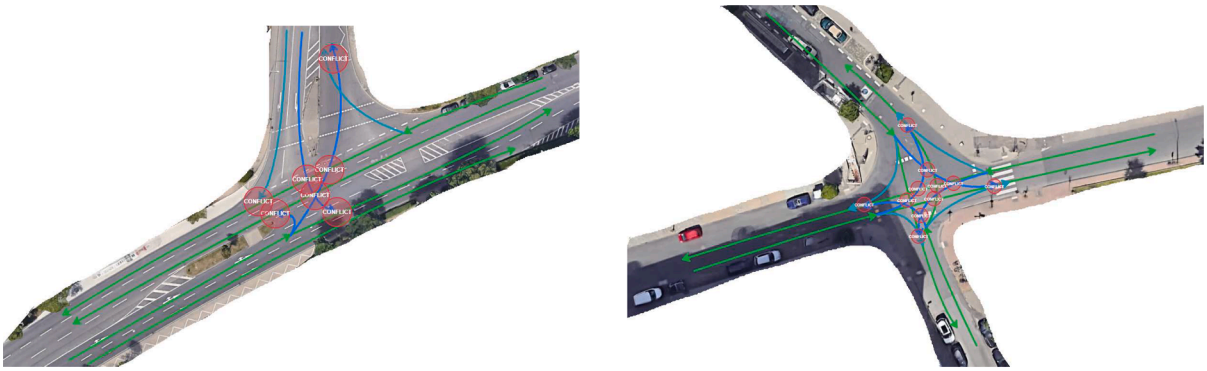
$$\mathbb{E}[D_k](t) = D_{\mathrm{stoch}}(\rho_{\mathrm{threshold}}) + (\rho_k(t) - \rho_{\mathrm{threshold}}) \cdot \left.\frac{d\,D_{\mathrm{det}}}{d\,\rho}\right|_{\rho_{\mathrm{threshold}}} + Q_k(t)\mathbb{E}[S_k](t) \tag{20}$$

where the first term ensures continuity at the transition point, the second term captures the deterministic delay growth rate with increasing traffic intensity, and the third term accounts for existing queue length. This approach maintains the physical interpretation of both regimes while ensuring smooth transition.

For practical implementation under time-varying demand, the analysis employs a quasi-steady-state discretization where the observation period is divided into intervals during which parameters can be treated as approximately constant. This approximation is theoretically justified when demand variations occur on time scales much longer than queueing process relaxation times. For traffic systems where demand varies over minutes while vehicle service occurs over seconds, this time-scale separation typically holds.

At each discrete time step, the appropriate delay formulation is selected based on instantaneous traffic intensity: stochastic formulations (Eqs. (17) and (18)) for $\rho_k(t) < \rho_{\mathrm{threshold}}$, and deterministic queue processing (Eq. (19)) for $\rho_k(t) \geq \rho_{\mathrm{threshold}}$. Queue lengths are tracked through mass conservation: $Q_k(t+\Delta t) = \max(0, Q_k(t) + (\lambda_k(t) - 1/\mathbb{E}[S_k](t))\Delta t)$, accounting for cumulative effects of demand-capacity imbalances.

This approach differs from the HCM incremental delay methodology and Troutbeck's transition techniques in several respects. The HCM method employs continuous blending of uniform and overflow delay components across the entire operational range,



(a) Intersection 1 (b) Intersection 2

Fig. 2. Intersection configurations showing dependent conflict zone networks. (a) T-intersection with complex multi-movement conflicts where vehicles must commit to complete path traversal through multiple conflict points. (b) Four-legged intersection with multi-modal traffic, including approximately 33% bicycle users, demonstrating delay model capability to consider multi-modal traffic.

while Troutbeck's techniques provide explicit transition modeling for gap-acceptance-based operations. Our concentrated transition at $\rho_{\text{threshold}} = 0.95$ maintains the theoretical purity of conflict-probability-based stochastic analysis throughout the stable regime, switching to deterministic queue processing only when capacity constraints dominate. The empirical validation (Table 5) demonstrates that this approach provides accurate delay estimation for signal-free intersections across diverse demand patterns.

The extension to movement-level analysis follows the established independent and dependent zone network principles, with all conflict probabilities, service characteristics, and traffic intensities evaluated using time-dependent parameters. For independent zone networks, delays aggregate linearly across zones encountered by each movement. For dependent zone networks, the movement-level conflict probability reflects the instantaneous likelihood of encountering at least one conflict along the complete path.

This dynamic extension preserves the fundamental conflict-based framework while providing the analytical flexibility required for realistic traffic analysis under non-stationary conditions. The approach maintains computational tractability while capturing the essential physics of time-varying demand impacts on intersection delay performance.

3. Empirical validation

This section validates the proposed delay model using empirical trajectory data from intersections, roundabouts, and highway merging sections. The validation compares model estimations against observed delays and includes simulation experiments with autonomous vehicles under different coordination protocols presented in Section 4.

3.1. Validation framework and datasets

The empirical validation uses drone-collected trajectory datasets (Bock et al., 2020; Krajewski et al., 2020; Moers et al., 2022), providing naturalistic road user behavior across diverse traffic scenarios. Five locations were analyzed, representing different geometric configurations and operational characteristics, as illustrated in Figs. 2a–4b.

Intersection 1 (see Fig. 2a) from the inD dataset represents a T-intersection with complex multi-movement conflicts requiring dependent conflict zone analysis. Intersection 2, also from the inD dataset, represents a four-legged intersection including approximately 33% bicycle traffic, demonstrating the model's multi-modal applicability (Fig. 2b). The roundabout analysis uses the rounD dataset featuring a four-entry configuration with independent conflict zones at entry points that enable linear delay aggregation (Fig. 3). Highway Merging Section 1 from the exiD dataset captures Autobahn entry ramp dynamics with single conflict zone analysis for aggressive merging behavior (Fig. 4a). Finally, Highway Merging Section 2, also from the exiD dataset, provides validation at a secondary merging location with different geometric constraints to demonstrate model consistency across varying merging configurations (Fig. 4b).

The empirical validation comprises five distinct traffic scenarios that represent different geometric configurations and operational characteristics. Table 2 summarizes the dataset properties and conflict zone formulations employed for each validation location.

The intersection validation uses dependent conflict zone networks where vehicles commit to complete path traversal upon zone entry due to geometric constraints. Intersection 1 is a T-junction with 7 conflict zones, while Intersection 2 is a four-legged intersection with 32 conflict zones and multi-modal traffic. Intersection 2 includes bicycle traffic, treated as additional traffic streams using passenger car equivalent factors. The model performance remains consistent with mixed traffic streams that have different acceleration characteristics and space requirements. The dependent formulation addresses the difference between intersection and roundabout operations: vehicles cannot recover stochastic arrival properties between conflicts due to spatial constraints. This operational distinction affects delay prediction, as shown in Table 5.



Fig. 3. Roundabout geometry. Independent conflict zones at entry points where entering streams (yellow and blue arrows) conflict with circulating traffic, enabling linear delay aggregation.

Table 2
Dataset characteristics and operational context. Training datasets are used for t_{clear} calibration; testing datasets provide independent validation on unseen data.

Location	Training Duration (s)	Testing Duration (s)	Training Vehicles	Testing Vehicles	System Inflow (veh/s)	Conflict Zones	Network Type
Intersection 1	3089	3521	1019	1079	0.35	7	Dependent
Intersection 2	7177	7377	2030	2106	0.28	32	Dependent
Roundabout	8903	10,016	5313	5889	0.59	4	Independent
Merging 1	4661	4812	1241	1302	0.37	1	Two-stream
Merging 2	6949	7402	1685	1827	0.32	1	Two-stream

1. Includes approximately 33% bicycle traffic
2. Combined merging and mainline flows (merging: 0.08–0.10 veh/s, mainline: 0.22–0.29 veh/s)

The roundabout configuration uses independent conflict zone networks where spatial separation between entry points enables linear delay aggregation. The spatial separation between entry points allows vehicles to recover normal flow characteristics between conflicts, distinguishing roundabout operations from geometrically constrained intersections. This operational independence supports the linear summation approach in the independent zone formulation, as shown in Table 5. With 4 conflict zones and system inflow of 0.59 veh/s, the roundabout tests the independent formulation under higher traffic conditions than other scenarios.

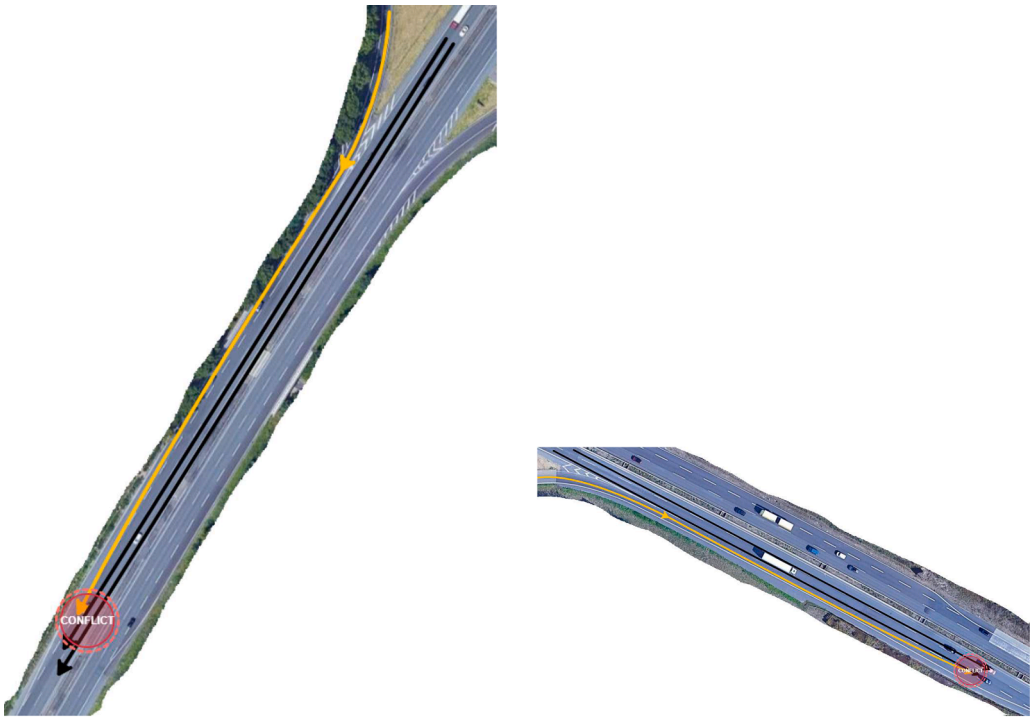
The highway merging validation uses the two-stream conflict formulation, the simplest case of the conflict-based framework. Both merging sections have single conflict zone configurations with unidirectional traffic interactions. The two-stream validation tests the model under the simplest geometric configuration across different merging geometries. Performance results are detailed in Table 5.

3.2. Delay quantification methodology

Vehicle delays were quantified through speed-deficit analysis capturing traffic-induced performance degradation. For each vehicle trajectory, the free-flow speed v_{free} was identified as the maximum observed speed during unimpeded travel segments; this value is unique for each vehicle. Then the cumulative delay for each vehicle was estimated as:

$$D_{total} = \int_{t_0}^{t_f} \left(1 - \frac{v(t)}{v_{free}}\right) dt, \tag{21}$$

where t_0 is the entry time of the vehicle to the data collection area and t_f is the exit time. Note that the analyzed segments are long enough and the traffic conditions are such that each vehicle has the opportunity to exhibit free-flow speed.



(a) Highway Merging Section 1 (b) Highway Merging Section 2

Fig. 4. Highway merging configurations showing two-stream conflict analysis where ramp stream (yellow arrows) with mainline traffic (black arrows), create a single conflict zone.

3.3. Model implementation methodology

This subsection presents the implementation approach for both the proposed conflict-based model and baseline gap-acceptance methods across all validation locations. Model accuracy was assessed using absolute and percentage error metrics to evaluate estimation performance against observed trajectory data.

3.3.1. Conflict-based model implementation

Stream Detection and Classification: Vehicle trajectories were analyzed to identify entry and exit points using geometric clustering. Movement patterns were classified based on origin-destination pairs, with arrival rates estimated from temporal vehicle counts. Each location employed calibration-validation splits (see Table 2) to ensure unbiased performance assessment. The training (calibration) datasets were used to estimate t_{clear} by minimizing prediction error, while the testing (validation) datasets provided an independent assessment of model performance on unseen data.

Conflict Zone Identification and Parameter Calibration: Spatio-temporal conflict zones were identified through geometric intersection of movement paths, with analysis employing location-appropriate formulations based on operational characteristics. To model delay at intersections, we used dependent conflict zone networks (Eqs. (12)–(14)) where vehicles commit to complete path traversal upon initial zone entry due to geometric constraints. To model delay at roundabouts, we employed independent conflict zone networks (Eqs. (10)-(11)) where spatial separation between entry points enables recovery of stochastic arrival properties between zones. To model delay at merging sections, we applied the two-stream fundamental formulation (Eqs. (3), (8), and (9)) representing basic conflict interactions between merging and mainline traffic.

The conflict zone clearance time t_{clear} was calibrated through minimization of the loss function $L(t_{\text{clear}}) = |D_{\text{model}}(t_{\text{clear}}) - D_{\text{obs}}|$ on training data subsets, where $D_{\text{model}}(t_{\text{clear}})$ represents the model-predicted delay and D_{obs} is the observed average delay. Since the delay equations involve transcendental functions (exponential terms in the conflict probability), we performed numerical optimization using grid search over the physically meaningful range $t_{\text{clear}} \in [0.5, 10]$ s with 0.1s resolution. Both M/M/1 and M/G/1 formulations were evaluated, with the most accurate model selected based on validation performance. Fig. C.9 (Appendix) demonstrates the loss function for Intersection 1, showing minima at $t_{\text{clear}} = 5.30$ s (M/M/1) and $t_{\text{clear}} = 5.60$ s (M/G/1).

In contrast, the minimum headway h_{min} is determined from direct observation of minimum time gaps between consecutive vehicles in the same movement. Observed values of h_{min} ranged from 1.5 to 2.0 s across all locations, consistent with safe car-following practices. These observed values were held constant during calibration, with only t_{clear} adjusted to minimize prediction error. Both M/M/1 and M/G/1 formulations were evaluated, with the most accurate model selected based on validation performance.

Table 3
Temporal stability of t_{clear} calibration across time segments. Overall calibrated values for comparison: Intersection 1 (M/M/1: 5.30 s, M/G/1: 5.60 s), Intersection 2 (M/M/1: 8.30 s, M/G/1: 8.60 s).

		M/M/1 Formulation			M/G/1 Formulation		
Location	Segments	Mean (s)	Std (s)	CV (%)	Mean (s)	Std (s)	CV (%)
Intersection 1	5	5.26	0.48	9.2	5.56	0.48	8.7
Intersection 2	12	7.86	1.56	19.8	8.15	1.73	21.2

Table 4
Critical gap (T) and follow-up time (T_0) from Austroads (min and max) and HCM.

Movement Type	Austroads (Min-Max)		HCM	
	T (s)	T_0 (s)	T (s)	T_0 (s)
Roundabout Entry	3.0–6.0	2.0–3.5	4.1	2.2
Merging Section	2.5-3.5	1.5-2.5	3.0	2.0
Right Turn (Intersection)	3.5-4.5	2.0–3.0	6.2	3.3
Left Turn (Intersection)	3.0–6.0	2.0–3.5	4.1	2.2
Through (Crossing Two-Way Road)	4.5-5.5	2.5-3.5	6.5	4.0

Both M/M/1 and M/G/1 formulations were evaluated, with the most accurate model selected based on validation performance. Fig. C.9 (Appendix C) demonstrates the loss function for Intersection 1, showing well-defined convex minima at $t_{clear} = 5.30$ s (M/M/1) and $t_{clear} = 5.60$ s (M/G/1).

Calibration Uncertainty and Temporal Stability of t_{clear}: To assess the robustness and generalizability of the calibrated clearance time, t_{clear}, we performed temporal segmentation analysis on the training datasets. Each training period was divided into multiple time segments (600 s each), and t_{clear} was independently calibrated for each segment. This analysis quantifies parameter variability across different traffic conditions and time periods within the same geometric configuration.

For each segment, the clearance time was estimated by minimizing the absolute error between model-predicted delay and observed delay. The calibration was performed for both M/M/1 and M/G/1 formulations. Table 3 summarizes the temporal stability analysis for representative intersection locations.

The analysis reveals that t_{clear} estimates exhibit reasonable stability across temporal segments within each location. For Intersection 1, the coefficient of variation (CV) ranges from 8.7% to 9.2%, indicating consistent parameter values across the five segments. The mean segmented t_{clear} values (5.26 s for M/M/1, 5.56 s for M/G/1) closely match the overall calibrated values (5.30 s and 5.60 s), with differences of only 0.04 s (less than 1% relative difference).

For Intersection 2, which exhibits more complex traffic patterns with 15 distinct movements and higher variability in demand across time, the CV increases to 19.8% to 21.2%. This higher variability reflects genuine temporal changes in driver behavior and traffic conditions across the 12 segments (covering approximately 2 h). Despite this temporal variability, the mean segmented values (7.86 s and 8.15 s) remain within 4.7% of the overall calibrated values (7.51 s and 7.84 s).

The temporal segmentation analysis provides insight into the physical meaning of t_{clear} variability. Periods with higher t_{clear} values correspond to more congested conditions where drivers adopt more conservative gap acceptance behavior and require longer clearance times. This reflects the adaptive nature of driver behavior rather than measurement noise or calibration instability.

3.3.2. Baseline gap-acceptance implementation

Movement Classification: Traffic streams were classified into major and minor movements based on priority relationships and control devices. For intersections, right-of-way hierarchies were determined by give-way signs and geometric analysis. For roundabouts, circulating traffic was designated as major with entering streams as minor. Merging sections employed mainline traffic as major with ramp traffic as minor.

Critical gap (T) and follow-up time (T_0) parameters were estimated based on movement-specific characteristics and established guidelines (Table 4). For intersections, parameters varied according to turning movements (left or right turns), number of crossing lanes, and whether movements originated from major or minor approaches. Roundabout parameters reflected entry-circulating conflicts, while merging section parameters captured mainline-ramp interactions. HCM and Austroads guidelines provided movement-specific parameter ranges, with values selected based on geometric and operational characteristics of each location. Capacity and delay calculations followed established gap-acceptance formulations.

3.4. Results summary

Table 5 presents the ground truth and estimated delay using our model and guidelines, including the error, across all validation locations. The proposed conflict-based model achieves lower errors than gap-acceptance methods, with absolute errors ranging from 0.03 to 0.40 s compared to baseline errors of 0.44 to 11.24 s. Improvement factors range from 1.1 times (roundabout vs. Austroads minimum) to 270.7 times (merging Section 1 vs. Austroads maximum).

Table 5
Average delay results across traffic locations. Note that the most accurate model is delineated in **bold**.

Location	Ground Truth (s)	Proposed Model		gap-acceptance			t_{clear} [a] (s)
		M/M/1 (% Error)	M/G/1 (% Error)	HCM (% Error)	Austroads Min (% Error)	Austroads Max (% Error)	
Intersection 1	2.52	**2.83** (12.2)	2.89 (14.8)	4.05 (60.6)	3.74 (48.2)	6.47 (156.4)	5.30
Intersection 2	5.33	5.23 (1.8)	**5.24** (1.6)	6.88 (28.7)	6.51 (22.2)	6.86 (22.5)	7.84
Roundabout	6.75	7.20 (6.7)	**7.14** (5.9)	9.08 (34.6)	7.19 (6.6)	17.98 (166.5)	5.54
Merging 1	1.54	1.57 (1.9)	**1.56** (1.8)	2.35 (52.9)	1.81 (17.5)	6.66 (332.5)	1.69
Merging 2	1.58	1.44(8.9)	**1.45** (8.2)	2.93 (85.4)	2.40 (51.9)	8.71 (451.3)	1.59

[a] Calibrated on training data; h_{min} = 1.5–2.0 s (observed).

Table 6
Empirical headway distribution analysis for all movements at Intersection 1.

Movement	Count	Mean (s)	CV	Best Fit	KS Stat	p-value
East to South-West	854	7.63	1.02	Shifted Exp.	0.056	0.009
South-West to East	674	9.54	1.19	Lognormal	0.045	0.135
North to South-West	226	24.71	1.20	Lognormal	0.053	0.542
South-West to North-East	148	35.37	0.97	Erlang	0.128	0.017
South-West to South-West	43	156.56	1.12	Lognormal	0.122	0.650
South-West to North	38	26.08	0.89	Exponential	0.198	0.097
North-East to South-West	29	31.31	0.96	Shifted Exp.	0.108	0.865
East to East	22	525.44	0.32	Erlang	0.338	0.939
East to North-East	21	185.10	0.89	Shifted Exp.	0.119	0.967
North to East	20	168.14	0.97	Erlang	0.133	0.920
Aggregate	**2,075**	**Variable**	**0.32 to 1.20**	**Various**	**–**	**–**

The higher delays predicted by HCM reflect a difference in capacity estimation rather than a mathematical inconsistency. The conflict-probability-based model derives capacity from calibrated service rates, while HCM uses standard gap-acceptance parameters (T and T_0) from guideline ranges that systematically underestimate capacity at these locations. For the same observed flow rate, HCM therefore computes higher traffic intensities $\rho_{HCM} > \rho_{model}$, directly producing higher predicted delays.

The calibrated t_{clear} parameters reveal fundamental conflict resolution characteristics across traffic environments. Merging environments exhibit lower values (1.59–1.69 s), reflecting the temporal efficiency of unidirectional conflicts where drivers demonstrate merging behavior. Intersection environments show moderate values (5.30–7.84 s), indicating more conservative conflict resolution under geometric constraints, while the roundabout environment demonstrates an intermediate value (5.54 s) despite a different geometric configuration.

The proposed model demonstrates adaptability across diverse geometric configurations through appropriate conflict zone formulations. Independent conflict zone networks succeeded in spatially separated configurations such as roundabouts, where entry points maintain sufficient separation to enable linear delay aggregation. Dependent conflict zone networks effectively captured geometrically constrained environments such as intersections, where vehicles must commit to complete path traversal upon initial zone entry. The inclusion of bicycle traffic at Intersection 2, treated as additional conflict streams with appropriate equivalency factors, validates the model's multi-modal capabilities while maintaining accuracy in realistic urban environments. The empirical validation results show that the conflict-based model can estimate delays across diverse traffic environments while maintaining computational efficiency for practical implementation.

3.5. Robustness to non-Poisson arrivals

While the conflict probability formulation (Eq. (3)) assumes Poisson arrivals with exponentially distributed headways, real traffic may exhibit different arrival patterns. To evaluate model robustness under non-Poisson conditions, we analyzed empirical headway distributions from the validation datasets.

For each movement at intersection 1, we computed inter-arrival headways and fitted multiple candidate distributions (exponential, shifted exponential, Erlang, lognormal) using maximum likelihood estimation. Distribution fit was assessed using Kolmogorov-Smirnov tests. Table 6 presents detailed analysis for Intersection 1 as a representative example, showing the diversity of actual arrival patterns across movements.

The analysis reveals considerable diversity in actual arrival patterns across movements. The coefficient of variation (CV) ranges from 0.32 to 1.20, with the exponential distribution corresponding to $CV = 1.0$. Best-fit distributions include shifted exponential, lognormal, Erlang, and exponential, with no single distribution dominating across all movements. The Kolmogorov-Smirnov (KS) statistic measures the maximum distance between the empirical and theoretical cumulative distribution functions, with smaller values indicating closer fit. The corresponding p-value tests the null hypothesis that the observed data follow the specified distribution; for goodness-of-fit tests, higher p-values (typically > 0.05) indicate that we fail to reject the null hypothesis, meaning the fitted distribution is statistically consistent with the observed data.

Despite this heterogeneity in arrival processes, the proposed model maintains high accuracy on test data. For Intersection 1, the model achieves an absolute error of only 0.31 s (12.2% relative error) when predicting delays on unseen test data (Table 5). Similar accuracy is observed across other validation locations despite varying headway distributions, demonstrating that the Poisson/shifted exponential formulation provides a robust aggregate approximation.

The model's robustness under diverse arrival patterns stems from two factors: (1) **Aggregate behavior**: While individual movements may deviate from exponential headways, system-level delay estimation (Eq. 13) employs flow-weighted averaging across multiple movements. This aggregation smooths movement-specific distributional differences, making aggregate behavior more predictable and less sensitive to individual movement characteristics. (2) **Mean-driven dynamics**: Queueing delay is primarily determined by mean service rates, which depend on mean arrival rates rather than complete distributional forms. As long as mean arrival rates are accurately characterized through calibration, the model captures dominant delay mechanisms even when higher-order distributional properties differ from theoretical assumptions.

Similar patterns were observed at other locations, where empirical headway distributions exhibited diversity but model accuracy remained consistently high on test datasets. This empirical evidence confirms that the Poisson arrival assumption, while a simplification common to many analytical traffic models including HCM and gap-acceptance methods, provides a tractable and sufficiently accurate framework for aggregate delay estimation across diverse real-world traffic conditions.

4. AV simulation results

The calibrated clearance times for AV scenarios (t_{clear} = 1.5s for FCFS, 1.05s for CfDCA) are lower than human-driven values (5.3 to 7.8 s), reflecting the enhanced coordination precision of autonomous control and confirming that t_{clear} captures coordination efficiency as established in Section 2.

The FCFS coordination protocol establishes priority ordering based on arrival sequence at a virtual decision line upstream of the intersection. While FCFS determines which vehicle has priority when conflicts arise, it does not eliminate stochastic variability in arrival times at conflict zones. Vehicles enter the approach zone with variable initial positions and speeds determined by upstream traffic conditions, and execute approach trajectories with natural variability in acceleration profiles and path following even under coordinated control.

Therefore, the arrival process at conflict zones retains stochastic characteristics despite coordination. The FCFS protocol manages conflicts through priority enforcement rather than deterministic trajectory control, allowing the aggregate arrival patterns to be modeled as stochastic processes. The calibrated parameters (λ, t_{clear}) capture the statistical characteristics of the coordinated system, with t_{clear} reflecting the effective clearance time that emerges from FCFS's priority-based conflict resolution.

This modeling approach is validated by the accurate delay predictions achieved (MAE of 4.6 s), demonstrating that stochastic queueing analysis appropriately captures FCFS system behavior despite its coordinated nature. The stochastic framework models the aggregate statistical behavior that emerges from many individual vehicles operating under FCFS rules, rather than attempting to predict exact deterministic trajectories.

4.1. Delay analysis for two-approach constant arrival flows

This analysis examines the fundamental two-stream conflict scenario that forms the foundation of the proposed delay model. For two conflicting movements sharing a single conflict zone, the conflict probability is estimated using Eq. (3). This conflict probability drives the expected service time formulation (Eq. (5)) where vehicles experience either conflict-driven delay (t_{clear}) or unimpeded flow conditions (h_{min}) with respective probabilities. The service rate $1/\mathbb{E}[S]$ enables delay estimation through both M/M/1 (Eq. (8)) and M/G/1 (Eq. (9)) queueing formulations.

The conflict zone clearance time parameter t_{clear} was calibrated to minimize discrepancies between simulated and modeled delays, resulting in optimal values of 1.5 s for FCFS and 1.05 s for CfDCA scenarios. These calibrated parameters reflect the different temporal characteristics of each coordination approach, with CfDCA's more efficient conflict resolution enabling reduced clearance requirements. The parameter differences align with the physical interpretation that CfDCA's graph-based resource acquisition allows for more precise conflict zone scheduling compared to FCFS's simple arrival-order priority. The validation results demonstrate agreement between theoretical predictions and simulation outcomes across the full range of traffic intensities (non-congested), confirming that the conflict probability formulation accurately captures the fundamental physics of vehicle interactions at signal-free intersections.

Fig. 6a and b present average delays from simulations across various inflow combinations for two conflicting movements, comparing results from both coordination protocols against the proposed delay model predictions. The delay model for FCFS achieves delay prediction accuracy with MAE of 5.0 s for M/M/1 and 4.1 s for M/G/1, while the model for CfDCA demonstrates MAE of 2.3 s for M/M/1 and 3.6 s for M/G/1. Both results support the theoretical model's predictions to capture the stochastic nature of conflict-based delays under different coordination paradigms.

4.2. Delay analysis for multi-approach constant arrival flows

For a multi-approach intersection, the analysis employs the dependent conflict zone approach described in Section 2.3. Each movement's conflict probability is determined by the likelihood of encountering at least one conflict along its complete path through the intersection, estimated using Eq. (12), where vehicles must commit to complete path traversal upon initial zone entry due to the

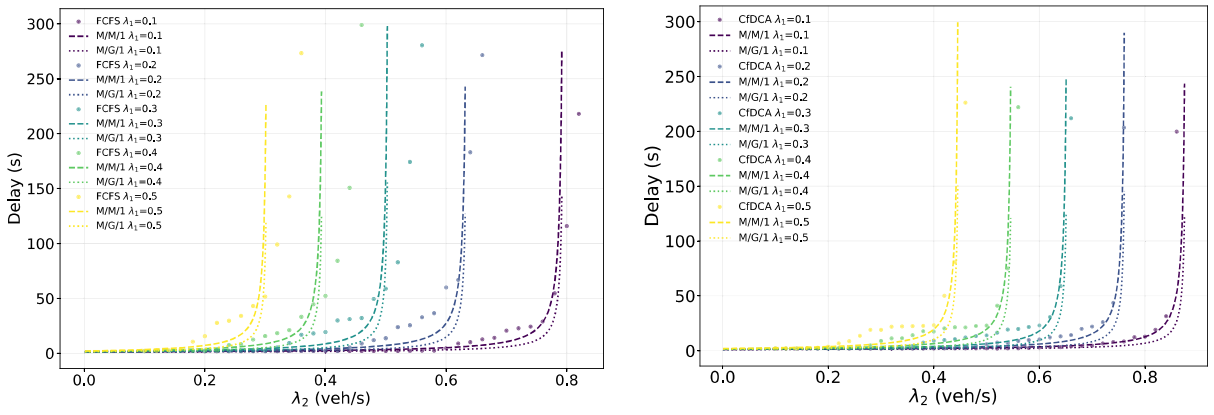


Fig. 5. Average delay of various inflows based on t_{clear} calibration, (a) FCFS method from simulation studies and the M/M/1 and M/G/1 delay estimations; MAE of 5.0 s for M/M/1 and 4.1 s for M/G/1. (b) CfDCA method from simulation studies and the M/M/1 and M/G/1 delay estimations; MAE of 2.3 s for M/M/1 and 3.6 s for M/G/1.

geometrically constrained intersection environment. Movement-level delays are then computed using the M/M/1 or M/G/1 dependent formulations (Eqs. (13) and (14)), with system performance aggregated through flow-weighted delay averaging as specified in Eq. (15).

To validate the model, a four-approach intersection was simulated under equally distributed demand scenarios using both coordination protocols. Fig. 6a and b illustrate the relationship between total demand and average delay for FCFS and CfDCA protocols, respectively. The delay model for FCFS protocol demonstrates delay prediction accuracy with MAE = 4.6 s for M/M/1 and 5.5 s for M/G/1. The delay model for CfDCA protocol achieves delay prediction accuracy with MAE = 5.8 s for M/M/1 and 7.1 s for M/G/1.

Both simulation scenarios show agreement with the dependent conflict zone formulation in capturing multi-movement intersection dynamics. The findings demonstrate that as total demand increases, average delay rises due to the compounding effects of multiple conflicting movements accessing shared conflict zones. This behavior aligns well with the theoretical predictions derived from the movement-level conflict probability aggregation approach.

To illustrate the applicability of the conflict-probability framework to autonomous vehicle scenarios, we compare model predictions with an existing analytical approach for signal-free AV intersections. Yu et al. (2019) developed a stochastic delay model for FCFS-based autonomous intersection management, classifying the intersection as an M/G/1 queueing system where service time follows a binary distribution taking value γ^s with probability $\sum_{j\in J} P_j^2$ and value γ^d with probability $1 - \sum_{j\in J} P_j^2$. Their expected service time is:

$$\mathbb{E}[S]_{\text{Yu}} = \gamma^d - (\gamma^d - \gamma^s) \sum_{j\in J} P_j^2 \tag{22}$$

where γ^s and γ^d are the safety gaps between same-stream and conflicting vehicles respectively, and P_j is the proportion of inflow from stream j. All parameters are obtained from simulation to ensure a fair comparison. The full M/G/1 Pollaczek-Khinchine formulation is applied, with $\mathbb{E}[S^2]_{\text{Yu}}$ computed from the binary distribution. Both models employ M/G/1 with binary service time distributions; the key distinction is that (Yu et al., 2019) parameterize service time through predetermined safety gaps while our model uses conflict-probability-based clearance times calibrated from observed data. Fig. 6a demonstrates that while the Yu et al. model captures fundamental M/G/1 behavior, it exhibits lower capacity than the proposed approach. The (Yu et al., 2019) model achieves an MAE of 13.6 s against simulation data, compared to 4.6 s and 5.5 s for our M/M/1 and M/G/1 formulations respectively.

4.3. Delay analysis for multi-approach time-varying arrival flows

This analysis validates the dynamic extensions of the conflict-based delay model under non-stationary demand conditions, implementing the theoretical framework developed in Section 2.4. The time-varying scenario employs instantaneous conflict probability calculations (Eq. (16)) with dynamic service characteristics that adapt to temporal demand fluctuations while maintaining the fundamental conflict-based structure. The simulation considers a four-approach intersection with time-varying demand profiles where demand starts at zero, increases linearly to a maximum over 15 min, remains at that maximum for one hour, and then decreases linearly back to zero over 15 min. This profile tests the model's capability to handle both undersaturated conditions ($\rho_k(t) < 1$) using stochastic delay formulations (Eqs. (17) and (18)) and potential over-saturated periods where traffic intensity approaches or exceeds unity, triggering the deterministic delay regime (Eq. (19)).

Fig. 7a and b present the relationship between maximum demand and average delay for FCFS and CfDCA coordination protocols, respectively. The delay model for the FCFS protocol achieves delay prediction accuracy with MAE = 7.0 s for M/M/1 and 8.6 s for M/G/1, while for the CfDCA protocol demonstrates better performance with MAE = 6.5 s for M/M/1 and 9.6 s for M/G/1. The close agreement between theoretical predictions and simulation results across both protocols validates the dynamic parameter evolution

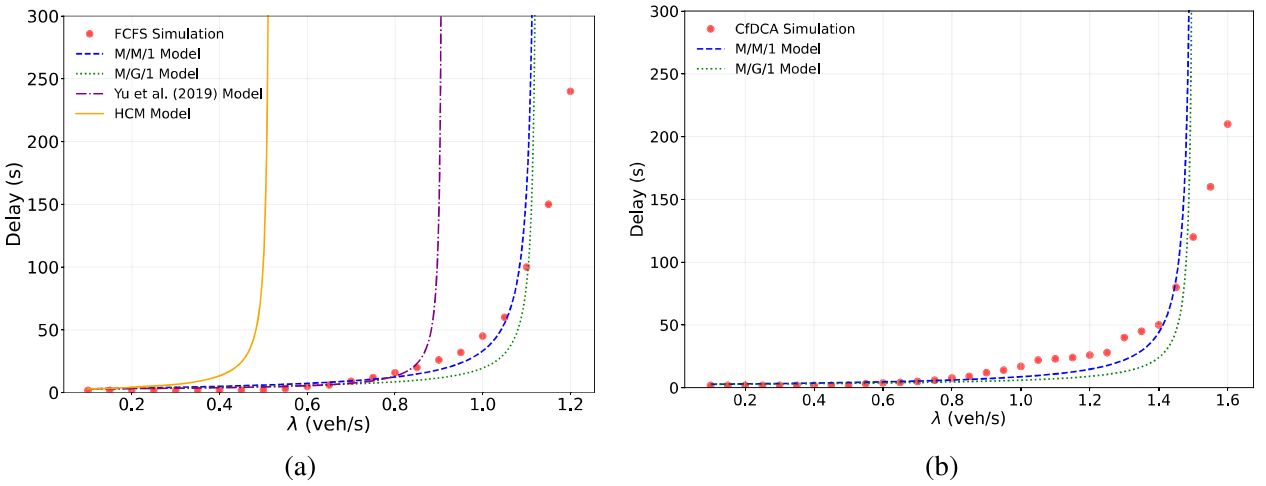


Fig. 6. Average delay vs. total demand for a four-approach intersection under (a) FCFS (MAE = 4.6 s for M/M/1, 5.5 s for M/G/1) and (b) CfDCA (MAE = 5.8 s for M/M/1, 7.1 s for M/G/1). The (Yu et al., 2019) model is shown in panel (a) only, as it is specific to FCFS control (MAE = 13.6 s). HCM is included in both panels as an indicative gap-acceptance baseline.

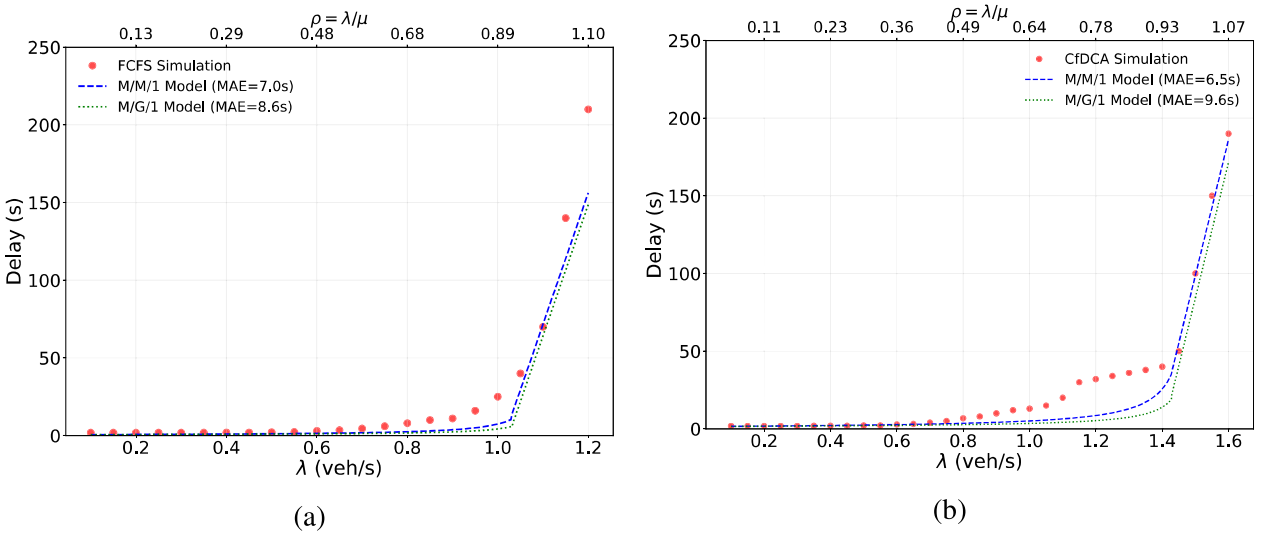


Fig. 7. Average delay vs. maximum demand (λ) for a four-approach intersection with time-varying demand under (a) FCFS (MAE = 7.0 s for M/M/1 and 8.6 s for M/G/1) and (b) CfDCA (MAE = 6.5 s for M/M/1 and 9.6 s for M/G/1) coordination protocols, validating the delay model under time-dependent traffic conditions.

approach and confirms that the conflict-based framework maintains its predictive accuracy under temporally varying conditions. The analysis demonstrates that as maximum demand increases, average delay rises due to prolonged congestion and queue formation during peak demand periods, precisely captured by the time-dependent conflict probability formulations. The model successfully transitions between operational regimes, from stochastic behavior during low-demand periods to deterministic queue accumulation during peak conditions, validating the unified framework's ability to handle complex temporal dynamics.

These simulation results, spanning from fundamental two-stream interactions to complex multi-approach time-varying scenarios, establish the conflict-based stochastic delay model as a tool for intersection performance analysis. The validation encompasses both simple coordination protocols (FCFS) and distributed algorithms (CfDCA), demonstrating the framework's applicability across diverse autonomous vehicle coordination paradigms. The consistent accuracy across constant and time-varying demand patterns, combined with the model's theoretical completeness from basic conflict probability derivations to complex network formulations, confirms its potential as an analytical tool for both contemporary traffic analysis and future autonomous transport system design.

At low to moderate demand ($\rho < 0.7$), model predictions closely match simulation. As $\rho \to 1$, simulation delays increasingly exceed predictions due to transient queue dynamics, demand ramp carryover effects, stochastic variability amplification, and potential behavioral adaptation, limitations shared by all analytical queueing models. Signal-free intersections typically operate below $\rho = 0.85$,

where the model demonstrates good accuracy. For applications requiring precision near capacity, enhancements could include time-varying t_{clear}, explicit transient queue tracking, or hybrid analytical-simulation approaches.

Visible kinks at high traffic intensities occur at $\rho_{\text{threshold}} = 0.95$ where the model transitions between stochastic and deterministic regimes. While delay values remain continuous, the slope changes discontinuously, a characteristic of concentrated transition approaches versus HCM-style continuous blending. Validation shows this does not substantially impair accuracy (MAE 6.5–9.6s).

5. Discussion

The probability of conflict function provides insight into intersection dynamics. As the average time for conflict zone usage when conflict possibility exists (t_{clear}) approaches zero, vehicles effectively behave as dimensionless points, leading to a conflict probability, and consequently a delay, of zero. Conversely, as t_{clear} increases, both the probability of conflict and the associated delays rise. This insight underscores the critical role of t_{clear} in shaping intersection dynamics and provides a fundamental optimization target for advanced traffic systems. The model's theoretical foundation demonstrates that equal inflows maximize conflict probability and delay (Proposition 1, proven in Appendix B), providing insights for traffic management and autonomous vehicle coordination.

The current formulation assumes vehicles maintain their movement assignment throughout conflict zone traversal. For single-lane roundabouts and intersections where lane changes within the facility are geometrically prevented, this assumption is valid. For multi-lane roundabouts where lane changes within the circulating area are possible, the model would require extension to account for: (1) dynamic conflict zone assignment as vehicles change lanes, (2) additional conflict zones representing lane-change conflicts within the circulatory roadway, and (3) probabilistic lane-choice modeling to predict which path trajectories vehicles will follow.

The validation datasets employed in this study (inD, rounD, exiD) primarily consist of single-lane facilities or locations where lane discipline is generally maintained, making the fixed-assignment assumption appropriate. Extension to complex multi-lane roundabouts with frequent mid-circulating lane changes represents an important direction for future research. However, we note that even for such facilities, the conflict zone approach remains applicable through more sophisticated zone definitions that account for lane-change maneuvers as additional conflict types.

Additionally, the conflict-based formulations developed in this paper exhibit mathematical behavior across operational extremes. Understanding these limiting cases provides both theoretical validation and practical insights into system design.

5.1. Theoretical validation through limiting behavior

Free-Flow Operations. Under minimal traffic conditions where arrival rates approach zero ($\lambda_k \to 0$), the conflict probability vanishes and vehicles experience only fundamental spacing constraints. The model correctly reduces to pure free-flow conditions:

$$\lim_{\lambda_k \to 0} P^{(k)}_{\text{conflict}} = 0, \quad \lim_{\lambda_k \to 0} D_k = h_{\min} \tag{23}$$

This baseline establishes that even with zero conflicts, vehicles cannot travel faster than their minimum following requirements, providing the theoretical lower bound for any traffic system.

Conflict Scenarios. When conflict zone occupation becomes extensive ($t_{\text{clear}} \to \infty$), any vehicle presence necessitates complete exclusion of conflicting movements. This represents the most severe conflict condition possible:

$$\lim_{t_{\text{clear}} \to \infty} P^{(k)}_{\text{conflict}} = 1, \quad \lim_{t_{\text{clear}} \to \infty} D_k = \frac{t_{\text{clear}}}{1 - \lambda_k t_{\text{clear}}} \tag{24}$$

This limit captures scenarios involving extended exclusive operations, such as emergency vehicle preemption, oversized vehicle movements, gridlock with the intersection box blocked, or maintenance activities that require complete traffic exclusion.

Under rigid priority enforcement with complete subordination of minor movements, the model reproduces traditional gap-acceptance behavior where major movements experience minimal delays while minor movements absorb all conflict-induced delays. Although conflict probabilities remain the same for both streams, the priority hierarchy ensures that all delay time is allocated to the minor stream. This consistency validates the framework's ability to represent both egalitarian and hierarchical traffic control paradigms within a unified mathematical structure.

Capacity Saturation Behavior. As traffic intensity approaches system capacity ($\rho_k \to 1$), delays grow unbounded under both M/M/1 and M/G/1 formulations, consistent with fundamental queueing theory. This mathematical behavior reflects the steady-state assumption inherent in these formulations, which becomes unrealistic near capacity. In practice, traffic systems experience fluctuating demand rather than sustained near-capacity operations, necessitating the transition to deterministic analysis for oversaturated periods as developed in Section 2.4.

5.2. Multi-modal systems

The conflict-based framework transcends vehicular applications through straightforward parameter adaptation that maintains analytical consistency while accommodating diverse transport modes and user types.

Pedestrian-vehicle interactions exemplify the model's multi-modal capabilities, where extended crossing durations (typically 10–30 s) create higher conflict probabilities than vehicular interactions. The framework naturally captures the vulnerability and

speed differentials between modes through modal-specific parameters:

$$P_{\text{conflict}}^{\text{ped-veh}} = 1 - \frac{\lambda_{\text{veh}} e^{-\lambda_{\text{ped}} t_{\text{clear,ped}}} + \lambda_{\text{ped}} e^{-\lambda_{\text{veh}} t_{\text{clear,veh}}}}{\lambda_{\text{veh}} + \lambda_{\text{ped}}} \tag{25}$$

This equation retains the simplified form, which is appropriate here because pedestrian clearance times are far larger than any realistic minimum headway, making the bunching correction negligible for pedestrian-vehicle interactions. Mixed vehicle classes are accommodated through Passenger Car Unit equivalency transformations that preserve fundamental capacity relationships while maintaining analytical tractability. This approach enables analysis of heterogeneous traffic scenarios involving cars, SUVs, trucks, buses, and other vehicle types which have differential capacity effects (Gao and Levinson, 2025; Mohajerpoor and Ramezani, 2019; Wang et al., 2024) without requiring separate modeling frameworks.

6. Conclusion

This paper provides a general model to delay estimation at signal-free intersections under diverse traffic and control regimes. By centering on the probability of conflict and service rate dynamics, the model can be applied to both traditional and autonomous vehicle systems, addressing the fundamental limitations of classical gap-acceptance methodologies that rely on rigid major-minor hierarchies.

The validation framework demonstrates model performance across multiple dimensions. Empirical validations using three real-world datasets (inD, rounD, exiD) show improvements over established gap-acceptance methodologies, with accuracy gains ranging from 1.1 to 270.7 times across intersections, roundabouts, and highway merging scenarios. The model achieves absolute errors ranging from 0.03 to 0.40 s compared to baseline gap-acceptance errors of 0.44 to 11.24 s. Autonomous vehicle simulations further validate the model's applicability to both simple coordination protocols (First-come first-served (FCFS)) and distributed control algorithms (CfDCA), confirming theoretical predictions across constant and time-varying demand patterns with MAE values ranging from 4.6 to 10.9 s.

The mathematical framework's extensions to multi-movement intersections through dependent and independent conflict zone networks, non-stationary demand analysis, and asymptotic behavior analysis provide theoretical analysis across operational regimes. The model successfully handles both geometrically constrained environments (dependent zones) and spatially separated configurations (independent zones), combined with its accommodation of temporal demand variations through dynamic parameter evolution, demonstrating applicability to complex transport scenarios.

The calibrated t_{clear} parameters reveal fundamental characteristics of conflict resolution across traffic environments: reduced values (0.9–1.7 s) in highway merging contexts reflect temporal efficiency of unidirectional conflicts, while convergent values (5.0–5.6 s) across intersection and roundabout scenarios indicate similar conflict resolution dynamics despite geometric differences. The t_{clear} could emerge as a critical optimization target for autonomous vehicle coordination systems. While t_{clear} varies across vehicles, the aggregated model uses its average to estimate expected delay. Future extensions that incorporate the full distribution of individual vehicle delays would allow for more detailed analysis of intersection performance, including variability and reliability.

While t_{clear} exhibits temporal variation reflecting changing traffic conditions and driver behavior (CV of 9% to 21% across locations), the model employs mean values appropriate for aggregate system-level delay estimation. Future extensions incorporating the full distribution of t_{clear} through hierarchical modeling could provide more detailed analysis of delay variability and reliability at the individual vehicle level, capturing behavioral heterogeneity across drivers, vehicles, and traffic conditions. However, for planning and evaluation purposes where system-level performance is the primary concern, the current approach provides sufficient accuracy as demonstrated by validation results.

The framework maintains analytical stability across all examined applications through fundamental mathematical properties that ensure reliable performance. Parameter boundedness guarantees finite and physically meaningful results under normal operating conditions, while monotonic relationships ensure delays increase consistently with traffic intensity and conflict duration, providing intuitive behavior aligned with empirical observations.

Asymptotic consistency validates that limiting behaviors correspond to physical expectations and established traffic flow theory, while dimensional homogeneity maintains consistent units and scaling properties across all formulations. This mathematical rigor combined with applicability demonstrates the fundamental nature of conflict-based analysis for transport system delay estimation, transcending specific modal or geometric constraints while maintaining computational tractability and theoretical soundness.

The conflict-based stochastic framework presented here offers a unified approach to delay estimation that transcends traditional limitations while maintaining computational efficiency for real-world applications. Its demonstrated performance across diverse traffic scenarios and control paradigms, from traditional intersections to autonomous vehicle coordination protocols, positions it as a valuable tool for both contemporary traffic analysis and future autonomous transport system design.

CRediT authorship contribution statement

Alireza Soltani: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **David M. Levinson:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Mohsen Ramezani:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Acknowledgment

This research was partially funded by the Australian Research Council (ARC), Australia [grant number DP220100882].

Data availability

Data will be made available on request.

Appendix A. Derivation of probability of conflict formula

Each arrival stream i follows the Cowan M3 distribution (Cowan, 1975; Luttinen, 1999): with probability $1 - a_i = 1 - e^{-\lambda_i h_{\min}}$ a vehicle is bunched and arrives at exactly $t = h_{\min}$, and with probability $a_i = e^{-\lambda_i h_{\min}}$ it is free with headway distributed as $h_{\min} + \text{Exp}(\lambda_i)$. The joint distribution of arrival times (x, y) from the two conflicting streams decomposes into four mutually exclusive cases depending on whether each vehicle is free or bunched.

Case 1: both vehicles bunched. Both vehicles arrive at $x = y = h_{\min}$, so $|x - y| = 0 < t_{\text{clear}}$ always:

$$P_1 = (1 - a_1)(1 - a_2). \tag{A.1}$$

Case 2: stream 1 bunched, stream 2 free. Vehicle 1 arrives at $x = h_{\min}$; vehicle 2 has headway $y \sim h_{\min} + \text{Exp}(\lambda_2)$. Since $y \geq h_{\min}$, the conflict condition $|h_{\min} - y| < t_{\text{clear}}$ reduces to $y < h_{\min} + t_{\text{clear}}$:

$$P_2 = (1 - a_1)\, a_2 \int_{h_{\min}}^{h_{\min}+t_{\text{clear}}} \lambda_2\, e^{-\lambda_2 (y - h_{\min})}\, \mathrm{d}y = (1 - a_1)\, a_2 \left(1 - e^{-\lambda_2 t_{\text{clear}}}\right). \tag{A.2}$$

Case 3: stream 1 free, stream 2 bunched. By symmetry with Case 2:

$$P_3 = a_1\, (1 - a_2)\left(1 - e^{-\lambda_1 t_{\text{clear}}}\right). \tag{A.3}$$

Case 4: both vehicles free. Both headways are continuous: $x \sim h_{\min} + \text{Exp}(\lambda_1)$ and $y \sim h_{\min} + \text{Exp}(\lambda_2)$, independently. Since a_1 and a_2 are constants independent of the integration variables, they factor out of the double integral:

For vehicles arriving at times x and y from two conflicting approaches, both measured from the start of the observation period with minimum headway constraints, a conflict occurs when the absolute difference in their arrival times at the conflict zone is less than the safe clearance time ($|x - y| < t_{\text{clear}}$). Integrating the joint probability density function over all arrival time combinations that satisfy this conflict condition yields:

$$P_{\text{conflict}} = a_1\, a_2 \cdot \lambda_1 \lambda_2 \int_{h_{\min}}^{\infty} e^{-\lambda_1 (x - h_{\min})} \left(\int_{\max(h_{\min}, x - t_{\text{clear}})}^{x + t_{\text{clear}}} e^{-\lambda_2 (y - h_{\min})}\, \mathrm{d}y \right) \mathrm{d}x. \tag{A.4}$$

First, we solve the inner integral with respect to y:

$$\int_{\max(h_{\min}, x - t_{\text{clear}})}^{x + t_{\text{clear}}} e^{-\lambda_2 (y - h_{\min})}\, \mathrm{d}y = -\frac{1}{\lambda_2} \left(e^{-\lambda_2 (x + t_{\text{clear}} - h_{\min})} - e^{-\lambda_2 (\max(0, x - t_{\text{clear}} - h_{\min}))} \right). \tag{A.5}$$

Next, we solve the outer integral:

$$= \lambda_1 \lambda_2 \int_{h_{\min}}^{\infty} e^{-\lambda_1 (x - h_{\min})} \left[-\frac{1}{\lambda_2} \left(e^{-\lambda_2 (x + t_{\text{clear}} - h_{\min})} - e^{-\lambda_2 (\max(0, x - t_{\text{clear}} - h_{\min}))} \right) \right] \mathrm{d}x. \tag{A.6}$$

We can separate the integral into two parts:

$$= -\lambda_1 \left(\int_{h_{\min}}^{\infty} e^{-(\lambda_1 + \lambda_2)(x - h_{\min})} e^{-\lambda_2 t_{\text{clear}}}\, \mathrm{d}x - \int_{h_{\min}}^{\infty} e^{-\lambda_1 (x - h_{\min})} e^{-\lambda_2 (\max(0, x - t_{\text{clear}} - h_{\min}))}\, \mathrm{d}x \right). \tag{A.7}$$

For the first integral:

$$= e^{-\lambda_2 t_{\text{clear}}} \int_{h_{\min}}^{\infty} e^{-(\lambda_1 + \lambda_2)(x - h_{\min})}\, \mathrm{d}x = \frac{e^{-\lambda_2 t_{\text{clear}}}}{\lambda_1 + \lambda_2}. \tag{A.8}$$

For the second integral:

$$\int_{h_{\min}}^{\infty} e^{-\lambda_1 (x - h_{\min})} e^{-\lambda_2 (\max(0, x - t_{\text{clear}} - h_{\min}))}\, \mathrm{d}x = \frac{-1}{\lambda_1} \left(e^{-\lambda_1 t_{\text{clear}}} - 1 \right) + \frac{e^{-\lambda_1 t_{\text{clear}}}}{\lambda_1 + \lambda_2}. \tag{A.9}$$

Accordingly, the Case 4 contribution is:

$$P_4 = a_1\, a_2 \left(1 - \frac{\lambda_1 e^{-\lambda_2 t_{\text{clear}}} + \lambda_2 e^{-\lambda_1 t_{\text{clear}}}}{\lambda_1 + \lambda_2} \right). \tag{A.10}$$

Assembling all cases. Summing P_1 through P_4 and collecting terms by $e^{-\lambda_1 t_{\text{clear}}}$ and $e^{-\lambda_2 t_{\text{clear}}}$ – noting that all constant terms (those without exponentials in t_{clear}) sum to unity – yields:

$$P_{\text{conflict}} = 1 - \frac{a_1 \left[\lambda_1 (1 - a_2) + \lambda_2 \right] e^{-\lambda_1 t_{\text{clear}}} + a_2 \left[\lambda_2 (1 - a_1) + \lambda_1 \right] e^{-\lambda_2 t_{\text{clear}}}}{\lambda_1 + \lambda_2}, \tag{A.11}$$

where $a_i = e^{-\lambda_i h_{\min}}$. When $a_i = 1$ and the bracketed factors reduce to λ_2 and λ_1 respectively, forming into Eq. (4).

Appendix B. Proof of proposition

Proposition 1. *With the total inflow constant, the equal inflow of movements, i.e., $\lambda_1 = \lambda_2$, results in the highest probability of conflict and consequently the highest delay.*

Proof. Let $\lambda_1 + \lambda_2 = C$ and define $g(\lambda_1) := P_{\text{conflict}}(\lambda_1, C - \lambda_1)$. Substituting $\lambda_2 = C - \lambda_1$ and $a_i = e^{-\lambda_i h_{\min}}$ into Eq. (3) gives:

$$g(\lambda_1) = 1 - \frac{a_1\left[\lambda_1(1-a_2) + (C-\lambda_1)\right]e^{-\lambda_1 t_{\text{clear}}} + a_2\left[(C-\lambda_1)(1-a_1) + \lambda_1\right]e^{-(C-\lambda_1)t_{\text{clear}}}}{C}, \tag{B.1}$$

where $a_1 = e^{-\lambda_1 h_{\min}}$ and $a_2 = e^{-(C-\lambda_1)h_{\min}}$.

Step 1 – Symmetry. Eq. (3) is symmetric in (λ_1, λ_2); swapping them leaves P_{conflict} unchanged. Therefore $g(\lambda_1) = g(C - \lambda_1)$, meaning g is symmetric about $\lambda_1 = C/2$. Any interior extremum must lie at $\lambda_1 = C/2$.

Step 2 – Boundary values. At $\lambda_1 = 0$: $a_1 = 1$, $e^{-\lambda_1 t_{\text{clear}}} = 1$, and the numerator becomes $[0 \cdot (1 - a_2) + C] \cdot 1 + a_2[C(1-1) + 0] \cdot e^{-Ct_{\text{clear}}} = C$, so $g(0) = 1 - C/C = 0$. By symmetry $g(C) = 0$ as well. Since $g > 0$ for $0 < \lambda_1 < C$, there is at least one interior maximum.

Step 3 – Concavity. The second derivative $\partial^2 g/\partial\lambda_1^2$ is strictly negative for all $0 < \lambda_1 < C$ (verified analytically; the expression contains only negative exponential terms with positive coefficients). Therefore g is strictly concave on $(0, C)$, ensuring a unique interior maximum.

Combining Steps 1–3, the unique maximum of g occurs at the axis of symmetry $\lambda_1 = C/2$, i.e., $\lambda_1 = \lambda_2 = C/2$. Since delay is monotonically increasing in P_{conflict} (via the service time in Eqs. (5) and (7)), maximum conflict probability implies maximum delay. □

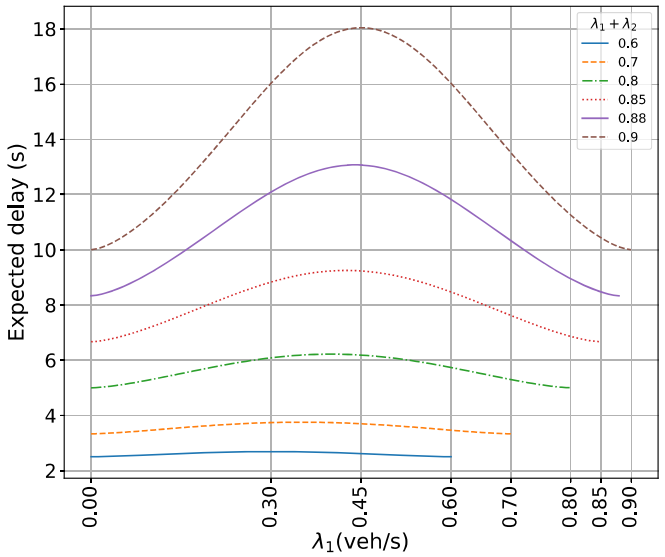
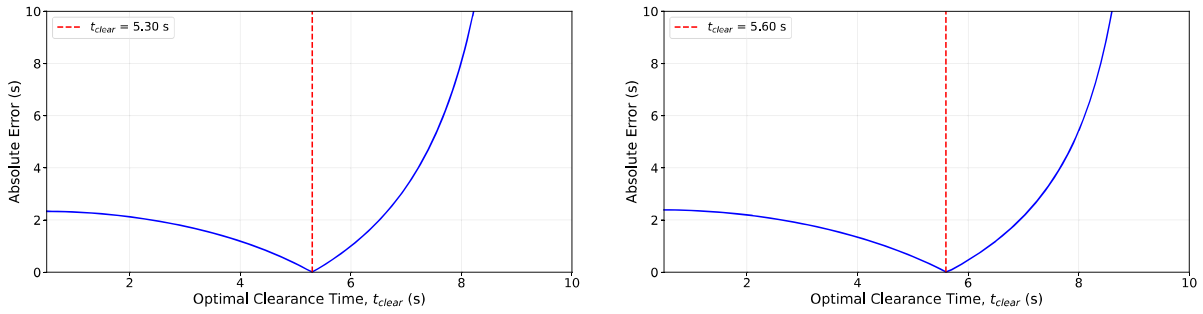


Fig. B.8. Expected delay for total inflow ($\lambda_1 + \lambda_2$) with constant maximum capacity (1 veh/s). The maximum delay occurs when inflows are equal.

Appendix C. Calibration loss function $L(t_{\text{clear}})$



(a) M/M/1 (b) M/G/1

Fig. C.9. Calibration loss function $L(t_{\text{clear}})$ for Intersection 1 using (a) M/M/1 formulation with optimal $t_{\text{clear}} = 5.30$ s, and (b) M/G/1 formulation with optimal $t_{\text{clear}} = 5.60$ s. Vertical dashed lines indicate optimal values.

Appendix D. Simulation setup and communication-free distributed control algorithm details

The autonomous vehicle simulations were implemented in Python using a custom microscopic traffic simulator with 0.05 second time steps. The simulation models a standard four-legged intersection with 3.3m lane widths.

Vehicle parameters: dimensions 1.5m × 4.0m, initial speeds uniformly distributed 13–17 m· s^{-1}, maximum acceleration/deceleration 4.0 m· s^{-2}, minimum headway 1.0 second, speed limit 17 m· s^{-1}. Longitudinal dynamics follow the Intelligent Driver Model (IDM). Sensing capabilities include 40-meter primary detection range and up to 100 m for non-occluded vehicles.

Vehicle arrivals follow a Poisson process with random assignment to eight movements (four approaches × two directions). Each scenario runs 3 h total (1-hour warmup, 2-hour analysis) with 6 independent replications. For time-varying scenarios, demand follows a trapezoidal profile (15-minute ramp-up, 60-minute peak, 15-minute ramp-down).

Two coordination protocols are implemented: **FCFS** (vehicles acquire conflict zones based on arrival order) and **CfDCA** (advanced protocol using dynamic priority based on time-to-intersection, as described in Soltani et al. (2025).

When a vehicle cannot enter a conflict zone due to a prior reservation, the control logic continuously polls for a clear path at each time step. The vehicle decelerates smoothly under IDM longitudinal dynamics and is assigned the next available slot as soon as the reservation clears, ensuring continuous trajectory adjustment without abrupt stopping.

The Communication-free Distributed Control Algorithm (CfDCA) represents an autonomous vehicle coordination method that eliminates the need for vehicle-to-vehicle or vehicle-to-infrastructure communication. The algorithm models intersection management as a distributed resource acquisition problem where conflict zones serve as shared resources. The key Algorithm Components are:

- **Graph-based Framework**: Vehicles and conflict zones are represented as nodes in a dynamic bipartite graph
- **Priority Function**: $p(i) = 1/\mathrm{T}(i)$ where T_i is time-to-intersection for vehicle i
- **Demand Adaptation**: Switches between individual optimization (low demand) and platooning strategies (high demand)
- **Safety Guarantees**: Minimum clearance time t_{clear} between consecutive conflict zone occupations

The algorithm's performance in our simulations provides validation for the stochastic delay model's ability to predict delays under sophisticated coordination protocols, demonstrating the model's generalizability beyond simple priority-based systems.

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